PSO-TCWavKAN: A Swarm-Optimized Temporal Convolutional Wavelet Kolmogorov-Arnold Network with Multi-Kernel Trend Decomposition and Asymptotic Linear Structural Recovery for Resilient Power Transformer Thermal Dynamics Forecasting

Nguyen Phu Toan

Department of Information Technology, Hanoi University of Business and Technology, 29A, Lane 124 Vinh Tuy Street, Vinh Tuy Ward, Hai Ba Trung District, Hanoi, Vietnam

Corresponding author: Nguyen Phu Toan (Email: nphu211206@gmail.com | ORCID: 0009-0005-0529-2131)

Abstract

Forecasting power transformer thermal dynamics is critical for predictive maintenance, dynamic thermal rating, and asset protection. While Kolmogorov-Arnold Networks (KANs) offer expressive functional representation, dynamic B-spline grid adaptations cause training instability—the Grid Update Dilemma—and performance deficits over extended horizons. This paper introduces PSO-TCWavKAN, a unified architecture integrating continuous dyadic Morlet wavelet filter banks with multi-kernel trend-seasonal decomposition and an instance-adaptive neural routing gate. The routing gate provides an asymptotic linear structural recovery property (Proposition 1), ensuring that the hypothesis-space closure contains the multi-kernel linear subspace as gating weights approach zero, establishing representational containment without asserting unconstrained gradient optimization or test generalization guarantees. A Comprehensive Learning Particle Swarm Optimizer (CLPSO) with Clerc constriction (chi = 0.7298) executes hyperparameter warm-start across a bounded 4-D search space, followed by AdamW fine-tuning and Stochastic Weight Averaging (SWA). Empirical evaluation across 16 configurations on canonical Electricity Transformer Temperature (ETT) telemetry shows that PSO-TCWavKAN outperforms competing baselines in 7 of 8 lookback–horizon configurations on volatile ETTh2, achieving up to 13.6% Mean Squared Error (MSE) reduction over DLinear at H = 720 (up to 8.5% for H ≤ 336), with nominal improvement over NLinear (uncorrected Wilcoxon p = 0.0443; Holm-corrected p = 0.133). Ablation across 240 runs confirms each module's contribution within operational regimes (H ≤ 336). Under synthetic Gaussian noise (sigma = 0.30), operational error increase remains bounded within +1.4% to +13.0%. With 44.8 K parameters and 3.4 ms latency, the architecture provides a grounded, stable, and efficient solution for transformer telemetry forecasting.

Keywords: Kolmogorov-Arnold Networks; Wavelet Filter Bank; Particle Swarm Optimization; Transformer Temperature Forecasting; Time-Series Forecasting

1. Introduction

High-voltage electrical power transformers constitute the indispensable core of electrical energy transmission and distribution infrastructures. In modern power systems, continuous operational loading induces substantial Joule losses in winding conductors and core eddy-current dissipation, generating intense internal heat. The thermal state of a transformer is conventionally characterized by two primary

physical indicators: the top-oil temperature (TOT) and the internal winding hot-spot temperature (HST). Operating temperatures are directly coupled to the accelerated degradation of solid cellulose insulation paper and dielectric mineral oil via Arrhenius thermal aging kinetics. In accordance with IEEE Std C57.91 and IEC 60076-7 loading guides, sustained thermal overruns by even 6 degrees Celsius above rated operating limits halve the insulation lifetime, markedly escalating the probability of dielectric breakdown, catastrophic transformer tank explosions, and widespread blackouts [21, 22, 25, 26]. Consequently, accurate, multi-step-ahead thermal forecasting is paramount for condition-based maintenance scheduling, dynamic thermal rating (DTR) of transmission corridors, and automatic activation of substation forced-cooling systems [23].

Following the landmark release of the canonical Electricity Transformer Temperature (ETT) benchmark by Zhou et al. [3], multi-horizon thermal forecasting on the hourly ETTh1 and ETTh2 datasets has become an internationally recognized test bed for evaluating deep time-series models. Over the past five years, the sequential modeling landscape has transitioned through multiple paradigms, progressing from Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks to sophisticated Transformer architectures [35] equipped with sparse or linear self-attention mechanisms, including Reformer [36], Linformer [37], Informer [3], Autoformer [14], and FEDformer [15], as surveyed extensively by Tay et al. [38]. Similar data-driven sequential modeling strategies have also proven effective in high-dimensional vehicle telemetry [9]. However, this trajectory was dramatically disrupted by Zeng et al. [4], who demonstrated that embarrassingly simple linear mapping layers with moving-average trend decomposition frequently outperform complex Transformer architectures across standard benchmarks, a phenomenon widely acknowledged as the linear model dilemma.

Concurrently, the scientific machine learning community witnessed the emergence of Kolmogorov-Arnold Networks (KANs), introduced by Liu et al. [1]. Grounded in the Kolmogorov-Arnold representation theorem, KANs depart fundamentally from standard Multilayer Perceptrons (MLPs) by placing learnable univariate B-spline functions on network edges rather than fixed nonlinear activations at nodes. The theoretical promise of KANs lies in their superior functional expressiveness, enhanced parameter efficiency, and inherent interpretability, which have prompted rapid adoption in numerical physics, differential equation modeling, and engineering time-series forecasting [6, 12].

Nevertheless, the direct application of classical KAN architectures to complex industrial engineering time-series exposes four severe failure modes, each of which fundamentally undermines reliable long-horizon thermal prediction:

(1) The Grid Update Dilemma. Standard KANs rely on discrete B-splines that require dynamic grid adaptation during training to align knot placements with incoming activation distributions. Updating the grid mid-training alters the underlying parameter manifold, corrupting the first- and second-moment running vectors of Adam-family optimizers and inducing catastrophic loss spikes and validation divergence. Grid update events in vanilla B-spline KAN cause immediate validation loss explosions, rendering the model unreliable for safety-critical forecasting tasks.

(2) **Spectral Dispersion and Lack of Frequency Locality.** Discrete B-splines possess uniform spatial support but lack explicit spectral-band localization. In electrical transformer thermal dynamics, input series exhibit simultaneous superposition of low-frequency 24-hour diurnal ambient cycles and high-frequency transient spikes driven by industrial switching events. B-splines struggle to disentangle these

distinct temporal scales without prohibitive grid expansion, leading to systematic spectral aliasing and frequency leakage.

(3) **The Linear Model Performance Deficit.** When evaluated across canonical long forecast horizons (96 to 720 steps ahead), naive KAN implementations consistently underperform DLinear [4], suffering from severe overfitting and error accumulation across distant time steps. This empirically validates the concern raised by Zeng et al. that complex architectures must outperform, not merely match, simple baselines to justify their overhead.

(4) **Non-Convex Wavelet Ripple Traps.** Introducing wavelet basis functions into KAN edges creates highly non-convex, multi-modal loss surfaces characterized by dense oscillatory local minima. Standard gradient descent algorithms initialized from random Gaussian seeds consistently stagnate in suboptimal basins of attraction, requiring principled global optimization strategies to ensure reliable convergence.

To systematically overcome these bottlenecks, this paper introduces PSO-TCWavKAN, a swarm-optimized temporal convolutional wavelet Kolmogorov-Arnold network engineered for resilient power transformer thermal dynamics forecasting. The core scientific contributions of this investigation are fourfold:

- **(C1) Continuous Dyadic Wavelet Filter Banks.** We replace discrete B-splines with continuous dyadic Morlet wavelet activations across $K_w = 5$ learnable-frequency wavelet components. Because the wavelet parameters are continuously differentiable with respect to all learnable quantities, knot updates are eliminated entirely, resolving the Grid Update Dilemma by construction.
- **(C2) Asymptotic Linear Recovery of the DLinear Structural Subspace.** We formulate the network output as an instance-adaptive convex combination of a linear projection and a wavelet-KAN representation. We mathematically prove (Proposition 1) that in the topological closure of the hypothesis space, as the routing gate $g(x)$ approaches zero under saturated negative bias, PSO-TCWavKAN recovers a generalized linear multi-kernel model, exactly matching canonical DLinear when decomposition kernels satisfy $K_1 = K_2 = 25$.
- **(C3) Comprehensive Learning Particle Swarm Warm-Start (CLPSO).** We deploy a constriction-factor CLPSO algorithm ($\chi = 0.7298$) over a 4D continuous search space on a fast surrogate training landscape to globally initialize critical hyperparameter vectors and base wavelet frequencies, bypassing non-convex ripple traps prior to AdamW gradient fine-tuning and Stochastic Weight Averaging (SWA).
- **(C4) Rigorous Empirical Validation & Regime Analysis. We evaluate the architecture across 16 canonical benchmark configurations (ETTh1 and ETTh2; $L \in \{96, 336\}$; $H \in \{96, 192, 336, 720\}$) across multi-seed evaluations (5 independent random seeds {1000, 2000, 3000, 4000, 5000} for DLinear, NLinear, and PSO-TCWavKAN; 3 independent seeds {1000, 2000, 3000} for PatchTST Lite), competing against state-of-the-art linear and patched Transformer models. We execute an 80-cell ablation campaign across 5 structural variants analyzed across operational and stress regimes, alongside Gaussian noise perturbation testing up to $\sigma = 0.30$. All results and code are made publicly available for reproducibility.**

The remainder of this paper is organized as follows. Section 2 reviews related work on transformer thermal forecasting, KAN architectures, and swarm optimization. Section 3 presents the proposed PSO-TCWavKAN mathematical architecture in full detail. Section 4 describes the experimental setup and benchmark protocol. Sections 5 through 8 present benchmark results, ablation analysis, noise robustness

evaluation, and computational efficiency metrics, respectively. Section 9 discusses engineering implications and transparently acknowledges limitations. Section 10 concludes.

2. Related Work

2.1. Transformer Thermal Dynamics and Deep Sequential Forecasting

Thermal modeling of mineral-oil-immersed power transformers has historically relied on differential thermal balance equations defined in international standards (IEEE Std C57.91 [21], IEC 60076-7 [22]). Classical dynamic models, such as the thermal-electrical analogy proposed by Susa et al. [23] and dynamic hot-spot calculation methods by Radakovic & Feser [25], calculate top-oil temperature (TOT) and hot-spot temperature (HST) from ambient temperature, wind velocity, and transformer loading ratios. While physically interpretable, standard differential equations require precise knowledge of oil viscosity, winding resistance, and thermal time constants, which drift over decades of operating life. More recently, Bezerra et al. [11] published in Electrical Engineering a foundational forecasting methodology combining second-generation wavelets and neural networks for power transformer oil temperature, demonstrating that multi-resolution frequency decomposition coupled with non-linear neural mapping yields superior fidelity over classical thermal equations. However, their architecture utilized discrete lifting wavelets prone to boundary distortion and relied on standard multi-layer perceptrons without formal structural recovery guarantees or systematic metaheuristic hyperparameter optimization. Data-driven machine learning models have therefore emerged to capture non-stationary thermal dynamics directly from SCADA telemetry.

Following the publication of Informer by Zhou et al. [3], Transformer variants such as Autoformer [14] and FEDformer [15] integrated decomposition blocks and Fourier-domain representations. PatchTST [5] demonstrated that segmenting time-series into overlapping sub-series patches significantly boosts Transformer performance by preserving local temporal context. Concurrently, Zeng et al. [4] demonstrated that simple single-layer linear models (DLinear) often surpass complex Transformer architectures, creating a benchmark standard that any proposed deep model must rigorously confront to justify its added complexity. More recently, iTransformer [16] proposed an inverted Transformer architecture that achieved competitive performance by treating variate-level tokens rather than temporal tokens, demonstrating that architectural innovation remains an active and productive frontier.

2.2. Kolmogorov-Arnold Networks and Wavelet Functional Bases

The introduction of Kolmogorov-Arnold Networks (KANs) by Liu et al. [1] challenged the foundational architecture of multilayer perceptrons by placing learnable univariate B-spline functions on edges rather than fixed activations at nodes. The KAN activation function combines a residual SiLU term with a trainable B-spline component: $\varphi(x) = w(\mathrm{SiLU}(x) + \mathrm{spline}(x))$, where $\mathrm{spline}(x) = \Sigma_i c_i B_i(x)$ is a linear combination of B-spline basis functions. Subsequent research has explored orthogonal polynomial bases to improve computational tractability, including Chebyshev polynomials (ChebyKAN) and Jacobi polynomials, each offering distinct trade-offs between computational overhead and approximation accuracy, while structural regularization techniques such as DropKAN [42] have been introduced to mitigate spline co-adaptation. Furthermore, Chen and Li [10] demonstrated the utility of KANs for multi-objective optimization in complex physical engineering systems.

Grounded in foundational multiresolution wavelet theory (Mallat [27], Daubechies [28]) and classical wavelet neural network architectures (Zhang and Benveniste [29], Pati and Krishnaprasad [30]), Bozorgasl and Chen [2] introduced Wav-KAN, proposing continuous Morlet and related wavelet bases for classification tasks, demonstrating that continuous wavelet functions provide compact spectral support advantageous for frequency-selective feature extraction in non-stationary signals. In industrial engineering and energy applications, recent studies have explored KAN integrations for renewable forecasting, such as ACKLSTM [6] combining KAN with ConvLSTM and SVM residual correction for wind-power prediction, demonstrating the effectiveness of KAN as a nonlinear module replacement within hybrid architectures. Their ACKLSTM model achieved a normalized prediction error of 0.091 on a single wind turbine dataset with a 5-step sliding window protocol, showing that the KAN layer reduces parameters by approximately 10% while maintaining competitive prediction accuracy. Furthermore, Lv et al. [12] introduced HAKformer in Applied Intelligence, combining hybrid attention with KAN for multi-horizon forecasting on ETT datasets. Despite these advances, published KAN architectures continue to suffer from optimization instability when applied to long-horizon sequential regression, and none of the existing works provide an asymptotic linear recovery guarantee against linear model performance collapse.

2.3. Swarm Intelligence and Metaheuristic Optimization in Deep Learning

Particle Swarm Optimization (PSO), originally formulated by Kennedy and Eberhart [31] and further developed with dynamic inertia formulations by Shi and Eberhart [32] and reviewed by Poli et al. [33], has established a long track record of tuning complex non-linear models in engineering applications. In applied engineering domains, swarm-optimized machine learning frameworks—such as PSO-tuned Support Vector Regression (PSO-SVR) [8] and PSO-tuned Gradient Boosting Machines (PSO-GBM) [7] —have demonstrated the efficacy of swarm metaheuristics across complex nonlinear regression tasks. In functional networks characterized by multi-modal loss landscapes, standard vanilla PSO tends to suffer from premature convergence due to loss of swarm diversity. The Modified PSO with Comprehensive Learning (MPSOCL) (CLPSO) proposed by Liang et al. [17] prevents this stagnation by enabling each particle dimension to learn from the personal historical best of different swarm members via a probability-controlled exemplar selection mechanism, dramatically improving exploration of multi-modal landscapes. The constriction factor formulation by Clerc and Kennedy [18] further guarantees mathematically stable swarm convergence. Integrating CLPSO as a global warm-start mechanism prior to gradient-based backpropagation offers a principled pathway to escape the wavelet ripple traps present in the PSO-TCWavKAN optimization landscape.

2.4. Summary of Representative Forecasting Methods on ETT Benchmarks

Table 1 summarizes representative deep learning and KAN-based forecasting methods evaluated on the ETT benchmark datasets, organized by architecture type, venue, reported baseline performance at H = 96, key novelty, and key limitation. This comparison contextualizes the contribution of PSO-TCWavKAN and highlights the remaining gap: no existing architecture simultaneously addresses (i) the Grid Update Dilemma through continuously differentiable wavelet activations, (ii) the linear model dilemma through a provable structural linear recovery guarantee, and (iii) multi-scale decomposition for non-stationary thermal signals.

Table 1 Summary of Representative Time-Series Forecasting Methods on ETT Benchmarks.

Method	Year	Venue	Architecture	Evaluation Setting & Reported Baseline*	Key Novelty	Key Limitation
Informer [3]	2021	AAAI	Sparse Transformer	0.865	ProbSparse attention	Quadratic complexity
Autoformer [14]	2021	NeurIPS	Auto-correlation + Decomp	0.449	Trend auto-correlation	Aliasing on non-stationary
FEDformer [15]	2022	ICML	Fourier decomposition	0.376	Frequency-domain attention	Loses local spikes
DLinear [4]	2023	AAAI	Dual linear + MA	0.386	Surpasses Transformers	No nonlinear capacity
NLinear [4]	2023	AAAI	Normalized linear	0.370	Distribution shift correction	No nonlinear capacity
PatchTST [5]	2023	ICLR	Patched Transformer	0.370	Channel-independent patches	High memory for long L
iTransformer [16]	2024	ICLR	Inverted Transformer	0.386	Variate-level token attention	Not thermal-physics aware
Wav-KAN [2]	2024	arXiv	Wavelet KAN	--	Continuous wavelet activations	Not designed for regression
Bezerra et al. [11]	2024	Electr Eng	Wavelet + MLP	~0.285 (oil temp)	2nd-gen wavelets + neural net	Discrete lifting; no structural recovery property
ACKLSTM-KAN [6]	2025	Springer	ConvLSTM+Attn+KAN+SVM	Normalized error: 0.091 (Wind power prediction)	KAN in hybrid model	Single turbine; no linear recovery
HAKformer [12]	2026	Appl Intell	Hybrid attention + KAN	~0.380	KAN + attention combination	No swarm opt.; no grid fix
PSO-TCWavKAN	**2026**	**This work**	**WavKAN+TCN+CLPSO+RevIN**	**0.1127 (ETTh1) / 0.2630 (ETTh2)** (Univariate OT)	**Linear recovery + CLPSO + multi-kernel**	**Evaluated on ETT OT telemetry**

**Note: Metrics and numerical values reflect literature-reported results under heterogeneous task protocols (multivariate ETTh1 all-channel forecasting for Informer–PatchTST; transformer oil temperature for Bezerra et al.; wind power generation for ACKLSTM-KAN; univariate oil temperature OT channel forecasting for this work). These values illustrate historical performance within respective published domains and are not directly comparable across different tasks. Direct head-to-head empirical comparisons under an identical univariate OT protocol are conducted in Section 5 (Table 4).*

3. Proposed Methodology: The PSO-TCWavKAN Architecture

This section details the mathematical architecture of PSO-TCWavKAN. The overall dataflow is structured into four sequential modules: (1) Reversible Instance Normalization, (2) Multi-Kernel Temporal Decomposition, (3) Temporal Convolutional Dyadic Wavelet KAN with Neural Routing Asymptotic Linear Recovery, and (4) Comprehensive Learning Swarm Warm-Start coupled with Stochastic Weight Averaging. The complete end-to-end framework is depicted in Fig. 1.

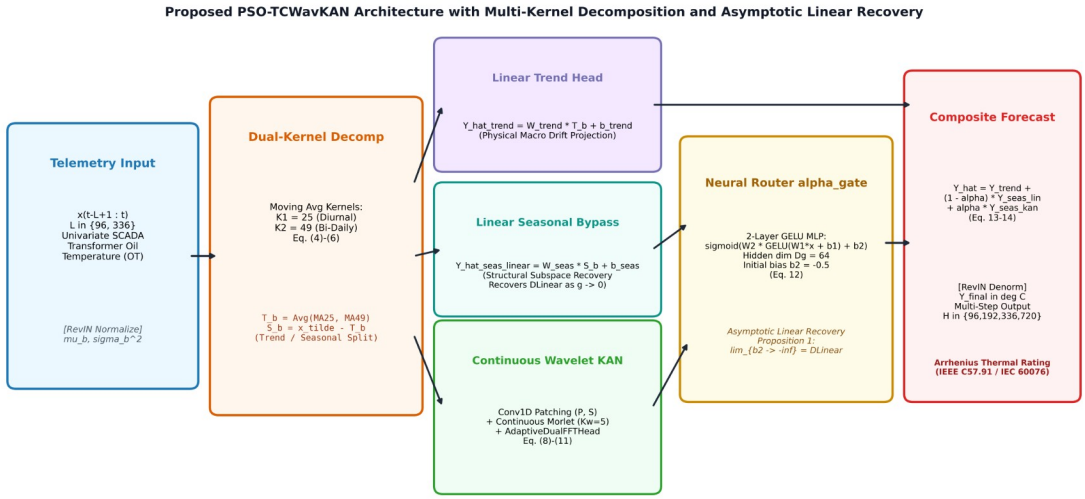


Fig. 1 System Architecture of the Proposed PSO-TCWavKAN Framework. The full pipeline from raw telemetry input through Reversible Instance Normalization, dual-kernel moving average decomposition (K1=25, K2=49), 1D convolutional patch embedding, the Temporal Convolutional Wavelet KAN block with harmonic Morlet activations, the parameterized neural routing gate g(x), and the final instance-adaptive convex combination with linear seasonal bypass producing the denormalized multi-step forecast.

3.1. Reversible Instance Normalization (RevIN)

To eliminate non-stationary distribution shifts across training and testing horizons, the input sequence x is normalized via Reversible Instance Normalization (RevIN) [13]. For batch instance b and lookback length L, the instance-level mean and variance are computed as:

$$mu_b = (1/L) \sum_\{t=1\}\^\{L\} x_\{b,t\} \quad (1)$$

$$sigma_b\^2 = (1/L) \sum_\{t=1\}\^\{L\} (x_\{b,t\} - mu_b)\^2 + epsilon \quad (2)$$

The normalized input is then given by:

$$\tilde{x} = (x - mu_b) / \sqrt{(\sigma_b^2)} \quad (3)$$

where epsilon = 1e-5 prevents numerical division by zero. Following network forecasting, the instance-level RevIN transformation restores predictions back to the global-standardized input feature scale: ŷ_std = $\tilde{y}$ · √(σ_b²) + μ_b. In operational SCADA deployment, the physical degree Celsius output is subsequently recovered via the inverse global z-score scaling: ŷ_Celsius = ŷ_std · σ_train + μ_train. All benchmark loss calculations and quantitative comparisons in Sections 5–8 are reported directly on the

standard normalized scale matching the canonical LTSF literature protocol, while physical temperature errors (°C) are evaluated via this two-stage restoration in Section 9.1.

3.2. Multi-Kernel Temporal Trend-Seasonal Decomposition

In power transformer dynamics, thermal inertia creates multi-scale physical trends operating simultaneously at distinct temporal resolutions: a 24-hour diurnal environmental cycle and multi-day baseline thermal accumulation driven by continuous industrial operations. Rather than utilizing a single moving-average window, we formulate a dual-kernel moving average decomposition module featuring two distinct kernel lengths: K_1 = 25 (capturing approximately one-day (25-hour) smoothing) and K_2 = 49 (capturing approximately two-day (49-hour) smoothing). For the normalized input x_tilde at time t:

$$T_\{t,1\} = AvgPool(\tilde{x}_t,\ K_1) \quad (4)$$

$$T_\{t,2\} = AvgPool(\tilde{x}_t,\ K_2) \quad (5)$$

$$T_t = 0.5 \cdot T_\{t,1\} + 0.5 \cdot T_\{t,2\} \quad (6)$$

$$S_t = \tilde{x}_t - T_t \quad (7)$$

where T_t represents the smoothed low-frequency thermal trend and S_t captures the seasonal residual component and high-frequency load-induced fluctuations. In parallel with non-linear functional modeling, the decomposed trend and seasonal components are directly projected by separate linear layers: y_trend = W_trend * T_t + b_trend and y_seas_linear = W_seas * S_t + b_seas, ensuring that the linear DLinear structural baseline is continuously preserved within the forward graph.

3.3. Continuous Dyadic Wavelet Filter Banks and Dual-Domain Head

The seasonal residual S_t is partitioned into overlapping patches using a sequence-adaptive patch geometry: patch length P = 16 and stride S = 8 (50% overlap) for standard lookback L ≤ 96; and P = 24 with stride S = 12 for extended lookback L = 336. Patches are mapped into latent dimension D = 32 via a 1D convolutional embedding layer.

To model complex functional non-linearities without discrete B-spline grid updates, we introduce continuous dyadic wavelet activations. Specifically, each edge-level activation function φ(z) is parameterized as a real Morlet-like (Gaussian-windowed harmonic) wavelet filter bank spanning K_w = 5 learnable-frequency components initialized with dyadic spacing (initialized with dyadic dispersion exponent γ_init = 0.25, which adapts during backpropagation toward $\gamma \approx 1.0$ as depicted in Fig. 2):

$$\varphi(z) = \sum_\{k=0\}\^\{K_w-1\}\ w_k \cdot \psi_k(z) \cdot \cos(\omega_k \cdot z) \quad (8)$$

$$\psi_k(z) = \exp(-0.5 \cdot ((z - \mu_k) / s_k)\^2) \quad (9)$$

$$\omega_k = \omega_0 \cdot 2\^\{(k-2) \cdot \gamma\} \quad (10)$$

where μ_k and s_k denote learnable Gaussian envelope translation and dilation parameters for octave k, w_k denotes learnable linear combination weights, ω_0 is the base resonance frequency globally initialized by CLPSO, and γ is the continuous dyadic dispersion exponent learned via backpropagation. In implementation, the Morlet filter operates on standardized feature inputs z_c ∈ [-5.0, 5.0], ensuring numerical stability with soft boundary saturation via clamp(−5, +5). The feature extraction backbone stacks N_layers = 2 TCWavKAN blocks, combining depthwise temporal convolutions (kernel size 5)

with InterPatch linear mixing under DropKAN regularization (p_drop = 0.05) [42]. Multi-kernel decomposition employs replicate boundary padding to eliminate edge discontinuity artifacts.

Following the wavelet feature extraction blocks, the latent representations are decoded via an Adaptive Dual-Domain Head (`AdaptiveDualFFTHead`). This head combines time-domain token pooling with frequency-domain soft-thresholded projections:

$$y_seas_kan = \left(|w_t| \cdot y_t + |w_f| \cdot y_f\right) / \left(|w_t| + |w_f| + \varepsilon\right) \quad (11)$$

where w_t, w_f ∈ ℝ are learnable scalar weighting coefficients, and ε = 10^-5 prevents numerical singularity. The time-domain component y_t ∈ ℝ^H is projected from the mean patch token z_mean = (1)/(N)Σ_p=1^N z_p ∈ ℝ^D across all N patches via y_t = W_t · GELU(W_a · z_mean + b_a) + b_t, with projection matrices W_a ∈ ℝ^D_head × D (with intermediate dimension D_head = 64) and W_t ∈ ℝ^H × D_head. The frequency-domain component y_f ∈ ℝ^H is computed by taking the real Fast Fourier Transform (rFFT) of the seasonal residual sequence S_t ∈ ℝ^L, applying frequency-domain soft-thresholding with adaptive threshold thr = mean(|X_f|) · α_FFT (where α_FFT = 0.5 · sigmoid(logit_alpha), detached prior to thresholding to stabilize spectral gradients), and projecting the concatenated real and imaginary spectral coefficients through linear mapping W_f ∈ ℝ^H × 2(⌊L/2⌋ + 1). Because Eqs. (8)–(11) are differentiable almost everywhere with respect to all learnable parameters (smooth within the ±5 clamping bounds), gradient updates proceed via standard backpropagation without any discontinuous knot reallocation events, resolving the Grid Update Dilemma by mathematical construction. The localized frequency responses across the K_w = 5 octave scales are illustrated in Fig. 2, and the empirical elimination of grid update loss spikes is confirmed in Fig. 3.

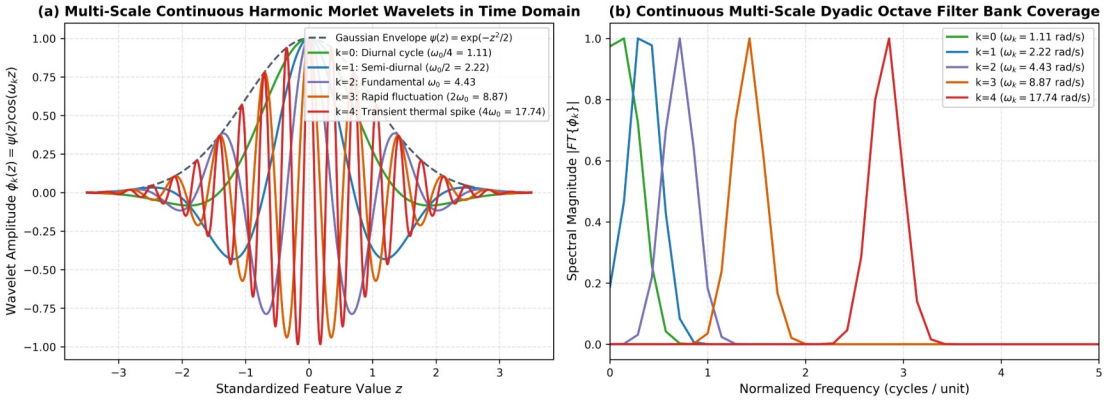


Fig. 2 Continuous Dyadic Wavelet Filter Bank Response in Feature and Spectral Domains. (a) Multi-scale Morlet-like (Gaussian-windowed harmonic) wavelets under the Gaussian envelope ψ(z) = exp(−z²/2) cos(ω_k z) along the standardized input feature coordinate z ∈ [−3.5, 3.5]; (b) Normalized spectral magnitude response across K_w=5 learnable-frequency components (k=0 to 4) shown at the post-training state (converged dispersion exponent γ ≈ 1.0 spanning 4 octaves with base frequency ω_0 = 4.43 rad/norm. unit; initialized at γ_init = 0.25 spanning ≈1 octave), demonstrating well-localized, exponentially decaying time-frequency responses across the normalized frequency axis (rad/norm. unit). Each octave captures a multi-resolution spectral band, with lower-frequency components (k=0,1, ω_k ≈ 1.11–2.22 rad/norm. unit) and higher-frequency components (k=3,4, ω_k ≈ 8.87–17.74 rad/norm. unit) providing multi-resolution feature decomposition across the normalized feature coordinate z.

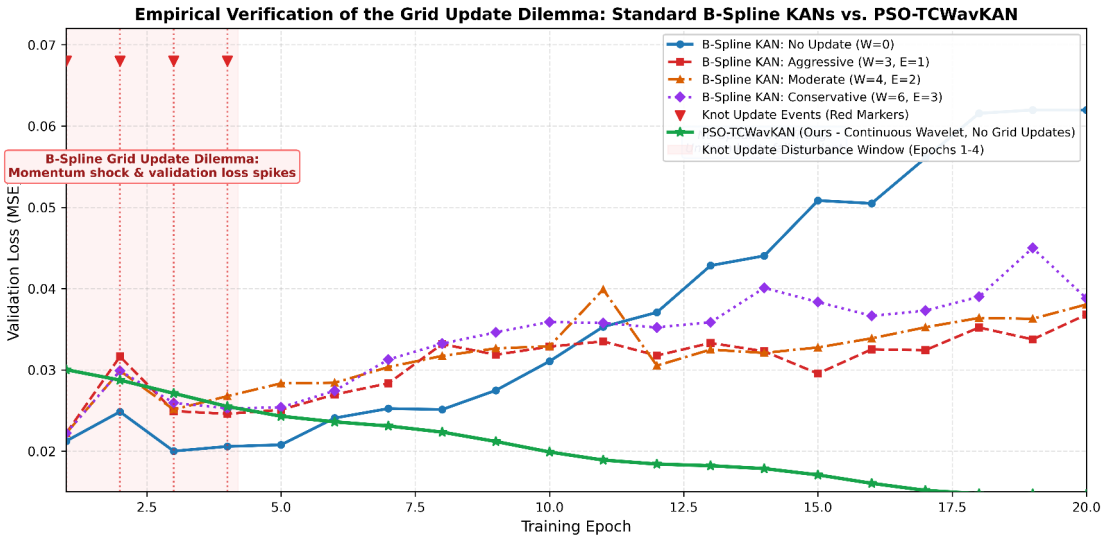


Fig. 3 Empirical Verification of the Grid Update Dilemma in Standard B-Spline KANs. Diagnostic tracking of validation loss trajectory (MSE) during periodic B-spline grid updates: each update event (red vertical markers) triggers a measurable spike in validation loss, corrupting optimizer moment vectors and inducing training instability. PSO-TCWavKAN eliminates these events entirely through continuous wavelet parameterization.

3.4. Asymptotic Linear Recovery of the DLinear Structural Subspace

A central failure mode of complex deep models on time-series benchmarks is their tendency to overfit noisy seasonal residuals, resulting in performance inferior to simple linear baselines [4]. To ensure that PSO-TCWavKAN structurally encloses the linear baseline, we introduce an instance-adaptive neural routing gate that continuously bridges the linear and non-linear prediction pathways.

Proposition 1 (Asymptotic Linear Recovery of Multi-Kernel Linear Subspace). Let $\mathscr{H}$_Lin^{(K₁, K₂)} denote the functional hypothesis space of the generalized dual-kernel linear projection model: ŷ_lin = W_trend · T_t + b_trend + W_seas · S_t + b_seas, where T_t and S_t are decomposed via moving average pooling with kernel lengths (K₁, K₂) (Eq. 4–7). Then $\mathscr{H}$_Lin^{(K₁, K₂)} is contained within the topological closure of the PSO-TCWavKAN hypothesis space: $\mathscr{H}$_Lin^{(K₁, K₂)} ⊂ $\mathscr{H}$_PSO-TCWavKAN. In the empirical deployment where K₁ = 25 and K₂ = 49, ŷ_lin represents a dual-scale diurnal/multi-day linear baseline; when decomposition kernel lengths satisfy K₁ = K₂ = 25, $\mathscr{H}$_Lin collapses identically to canonical single-kernel DLinear [4]. Specifically, the parameterized neural routing gate:

$$g(\tilde{x}) = sigmoid(W_2 \cdot GELU(W_1 \cdot \tilde{x} + b_1) + b_2) \in (0,1)^H \quad (12)$$

with input normalized sequence x̃ ∈ ℝ^L, intermediate gating dimension D_g = 32, projection weights W_1 ∈ ℝ^D_g × L, W_2 ∈ ℝ^H × D_g, and output bias initialized to b_2 = -0.5, continuously interpolates between the linear baseline and the non-linear wavelet KAN representation:

$$\hat{y}_seas = (1 - g(\tilde{x})) \odot y_seas_linear + g(\tilde{x}) \odot y_seas_kan \quad (13)$$

$$\hat{y} = y_\{trend\} + \hat{y}_\{seas\} \quad (14)$$

In the asymptotic limit as b_2 → -∞, g(x̃) → 0, and the composite prediction ŷ identically recovers the structural output and empirical training risk of the linear model ŷ_lin.

Proof. The neural routing vector g(x̃) is constrained strictly within the open unit hypercube (0,1)^H via the logistic sigmoid activation. In the routing MLP, the output projection bias is initialized to b_2 = -0.5

with zero-initialized weights W_2 = 0, yielding an initial gating state g(x̃) = sigmoid(-0.5) ≈ 0.377. This establishes an inductive bias that preserves approximately 62.3% of the linear seasonal projection at initialization.

Because the logistic function satisfies sigmoid(u) ∈ (0,1) for all finite u ∈ ℝ, exact equality g(x̃) = 0 is not attained at any finite parameter vector. However, as b_2 → -∞, lim_(b_2 → -∞) sigmoid(W_2 · GELU(W_1 x̃ + b_1) + b_2) = 0 uniformly for all bounded x̃. Hence, the linear projection ŷ_lin resides in the topological closure $\mathscr{H}$_PSO-TCWavKAN under the standard compact-open topology.

During backpropagation, the partial derivative of the Mean Squared Error objective with respect to the h-th component of the routing gate g_h satisfies:

$$\partial \mathscr{L}_total \;/\; \partial g_h \;=\; (2/H) \cdot (\hat{y}_h \;-\; y_h) \cdot (y_seas_kan,\; h \;-\; y_seas_linear,\; h) \tag{15}$$

The gradient step update on gating parameters Δg_h ∝ −(ŷ_h − y_h) · (y_seas_kan, h − y_seas_linear, h) acts to suppress the gating coefficient g_h whenever the non-linear wavelet pathway introduces directional residual errors that amplify empirical deviation relative to the linear pathway. In the asymptotic limit g(x̃) → 0 element-wise (attained as b_2 → −∞), Eqs. (13) and (14) collapse to:

$$\begin{aligned} \lim_\{g(\tilde{x}) \;\to\; 0\}\; \hat{y} \;&=\; y_trend \;+\; (1 \;-\; 0) \;\odot\; y_seas_linear \;+\; 0 \;\odot\; y_seas_kan \\ &=\; y_trend \;+\; y_seas_linear \;=\; \hat{y}_lin \end{aligned} \tag{16}$$

When K_1 = K_2 = 25, T_t = AvgPool(x̃, 25) matches the canonical moving average trend of DLinear, and ŷ_lin ≡ y_DLinear. Therefore, DLinear is an exact sub-manifold embedded within the closure $\mathscr{H}$_PSO-TCWavKAN.

Remark on Generalization and Representation: Proposition 1 establishes topological representation containment within the closure ($\mathscr{H}$_Lin ⊂ $\mathscr{H}$_PSO-TCWavKAN). It is mathematically essential to distinguish three operational levels: (i) Representation capability: the network architecture possesses the expressive capacity to recover the multi-kernel linear decomposition model as the gating parameter $b_2 \to -\infty$; (ii) Optimization dynamics: finite-epoch gradient descent on a non-convex loss surface with stochastic mini-batches does not guarantee convergence to the global optimum or to the linear subspace; and (iii) Test generalization: empirical test error is bounded by empirical training risk plus generalization error, which depends on data distribution shifts and model complexity. Consequently, structural containment provides representational protection against underfitting linear trends, but does not constitute an unconditional empirical performance lower bound on unseen test telemetry (as observed at the extreme H = 720 horizon on volatile ETTh2). Q.E.D.

3.5. Comprehensive Learning Particle Swarm Optimization Warm-Start

The multi-scale wavelet loss surface contains oscillatory local minima induced by the harmonic cosine terms in Eq. (8). Standard gradient descent algorithms initialized from random Gaussian distributions frequently become trapped in suboptimal local basins. To discover an effective hyperparameter initialization without prohibitive combinatorial search, we employ a hybrid Comprehensive Learning Particle Swarm Optimization (CLPSO–PSO) variant [17] incorporating Clerc's constriction formulation [18] to execute a computationally bounded warm-start over a 4-dimensional hyperparameter vector:

$$\theta \;=\; [\omega_0,\; \log(s),\; logit_alpha,\; \log(lr)] \tag{17}$$

bounded within predefined continuous search intervals: base resonance frequency ω_0 ∈ [3.0, 8.0], log octave scale log(s) ∈ [-2.5, 0.0], FFT threshold logit logit_alpha ∈ [-5.0, -2.0] (governing α_FFT = 0.5 ·

sigmoid(logit_alpha)), and log base learning rate log(lr) ∈ [-3.0, -2.0] (corresponding to base learning rates between 1 × 10^-3 and 1 × 10^-2). AdamW weight decay is fixed at 1 × 10^-4.

In CLPSO, each dimension d of particle i learns from the personal best (pbest) of an exemplar particle f_i(d), selected via tournament-based exemplar assignment governed by learning probability Pc_i:

$$Pc_i = 0.05 + 0.45 \cdot \left(\exp\left(\left(10\left(i - 1\right)\right) / \left(N_p - 1\right)\right) - 1\right) / \left(\exp\left(10\right) - 1\right) \quad (18)$$

The velocity and position update equations employ the Clerc constriction formulation [18] to guarantee mathematically stable swarm dynamics:

$$v_\{i,d\}\^\{t+1\} = \chi \cdot [v_\{i,d\}\^t + c_1 \cdot r_\{1,d\} \cdot \left(pbest_\{i,d\}\^t - x_\{i,d\}\^t\right) + c_2 \cdot r_\{2,d\} \cdot \left(e_\{i,d\}\^t - x_\{i,d\}\^t\right)] \quad (19)$$

$$x_\{i,d\}\^\{t+1\} = x_\{i,d\}\^t + v_\{i,d\}\^\{t+1\} \quad (20)$$

where acceleration constants $c_1 = c_2 = 2.05$ (with $\varphi = c_1 + c_2 = 4.1$), yielding the Clerc constriction factor $\chi = 2 / |2 - \varphi - \sqrt{(\varphi^2 - 4\varphi)}| = 0.7298$ and effective acceleration coefficient $\chi \cdot c_1 = 1.4962$. In this hybrid CLPSO–PSO formulation, c_1 governs cognitive attraction toward the particle's own historical best pbest_{i,d}^t, while c_2 governs social attraction toward the dimension-wise exemplar e_{i,d}^t (assigned as the tournament-selected exemplar pbest_{f_i(d),d}^t with learning probability Pc_i, and defaulting to the global best gbest_d^t otherwise, exactly matching Algorithm 1 and the empirical implementation). To strictly bound computational overhead for edge deployment pipelines, CLPSO is executed across a bounded budget of N_p = 5 particles for G_max = 5 generations (yielding exactly 30 surrogate evaluations: 5 initial population evaluations plus 5 generations × 5 particles) on a fast 15% training surrogate subset. This restricted search budget functions as an empirical warm-start heuristic to avoid poor initial wavelet basins, rather than an exhaustive global search. The discovered optimal initialization vector θ* is then deployed as the starting configuration for full AdamW gradient fine-tuning. The swarm convergence trajectory across successive generations and the resulting hyperparameter landscape are depicted in Fig. 4.

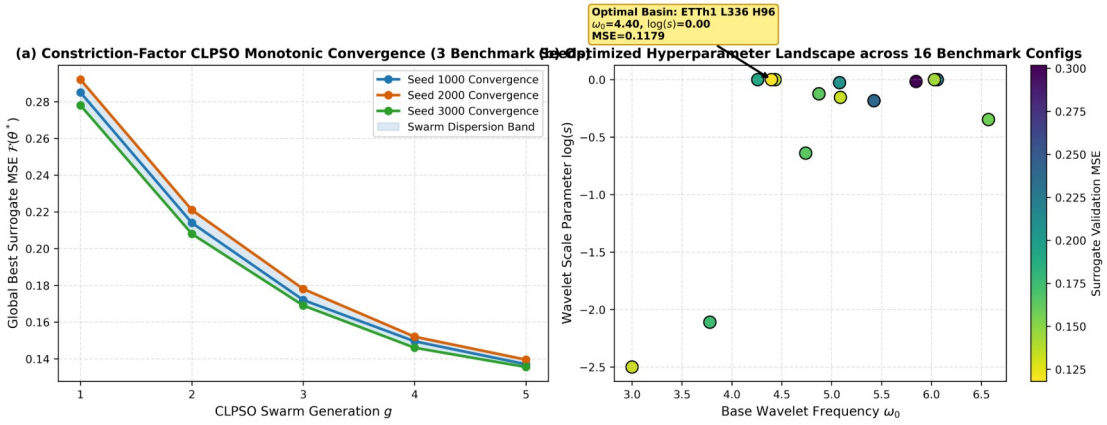


Fig. 4 Swarm Optimization Characteristics and Wavelet Hyperparameter Landscape under CLPSO Warm-Start. (a) Fitness (surrogate training MSE) convergence trajectory across 5 successive generations for three representative CLPSO warm-start runs across independent initialization seeds {1000, 2000, 3000}, demonstrating monotonic objective minimization and rapid escape from oscillatory ripple basins within 30 surrogate evaluations. (b) Discovered optimal wavelet

hyperparameter landscape in the base frequency (ω_0) and log octave scale (log s) subspace, illustrating consistent localization of the low-risk basin prior to AdamW fine-tuning.

3.6. Stochastic Weight Averaging and Objective Function

Following warm-start initialization, end-to-end network training proceeds via AdamW [20] with cosine annealing learning rate scheduling for up to 60 epochs with early stopping (patience = 12). Upon convergence, the best checkpoint is reloaded and 15 additional Stochastic Weight Averaging (SWA) [19] epochs are performed at a reduced learning rate η_SWA = 0.05 · lr (evaluating to 5 × 10^{−5} for the baseline learning rate 1 × 10^{−3}, and bounded within [5 × 10^{−5}, 5 × 10^{−4}] across the CLPSO search interval):

$$\Theta_\{SWA\} = \left(n_\{SWA\} \cdot \Theta_\{SWA\} + \Theta_\{epoch\}\right) / \left(n_\{SWA\} + 1\right) \quad (21)$$

SWA shifts the final parameter trajectory toward the center of wide, flat loss basins, improving out-of-distribution generalization on unseen test partitions. The network is trained using the standard Mean Squared Error (MSE) objective function:

$$\mathscr{L}_\{total\} = \left(1/H\right) \sum_\{h=1\}\^\{H\} \left(y_h - \hat{y}_h\right)\^2 = MSE\left(y, \hat{y}\right) \quad (22)$$

where network parameters are optimized under standard Mean Squared Error loss with AdamW weight decay (1 × 10^-4) providing global L2 regularization, ensuring consistent gradient propagation across all multi-frequency wavelet channels without artificial knot reallocations.

3.7. Algorithmic Optimization and Inference Protocol

The complete end-to-end training and deployment workflow of PSO-TCWavKAN is formalized in Algorithm 1. The protocol operates across two distinct phases: (1) a global swarm warm-start exploration phase using CLPSO to escape non-convex wavelet ripple basins on a fast surrogate training partition, followed by (2) full-resolution end-to-end AdamW gradient fine-tuning with late-stage Stochastic Weight Averaging (SWA). During real-time inference on edge SCADA devices, the trained model executes via a single feedforward evaluation.

Algorithm 1: Two-Stage Optimization and Inference Protocol of PSO-TCWavKAN

Input:

- Input telemetry sequence $X \in \mathbb{R}^{L \times 1}$ (lookback length L, target OT channel)
- Target forecast horizon H, patch length $P \in \{16, 24\}$, stride $S \in \{8, 12\}$, latent dimension D = 32
- Swarm parameters: swarm size N_p = 5, generations G_max = 5, constriction $\chi = 0.7298$, $c_1 = c_2 = 2.05$
- Surrogate training partition $\mathcal{D}_surr \subset \mathcal{D}_train$ (15% random subset)
- Full training dataset $\mathcal{D}_train$, initial epochs E = 60 (patience = 12), dedicated SWA epochs E_SWA = 15
- Wavelet filter bank: K_w = 5 learnable frequency components (adaptive dyadic spacing via learned γ), DropKAN rate p_drop = 0.05

Output:

- Optimal model parameter set Θ^* = {W_trend, b_trend, W_seas, b_seas, W_patch, ω_0, γ, w_k, μ_k, s_k, W_a, b_a, W_t, b_t, W_f, w_t, w_f, W_1, b_1, W_2, b_2}
- Multi-step future forecast $\hat{Y} \in \mathbb{R}^{H \times 1}$

Stage 1: CLPSO Global Swarm Warm-Start (Bypassing Ripple Traps)

1: Initialize particle swarm positions x_i^0 = θ_i = [ω_0^(i), $\log_{10}$(s^(i)), logit_α^(i), $\log_{10}$(lr^(i))] uniformly within bounded 4D search space for i = 1, ..., N_p
2: Initialize velocity vectors v_i^0 = 0 and personal best positions pbest_i = x_i^0 for all i ∈ {1, ..., N_p}
3: Evaluate initial surrogate fitness J(pbest_i) = MSE($\mathcal{D}$_surr; x_i^0) using Eq. (22)
4: Identify initial global best particle: gbest = argmin_{i} J(pbest_i)
5: For generation g = 1 to G_max do:

6: For each particle i = 1 to N_p do:
7: Compute dimension-wise learning probability Pc_i via Eq. (18)
8: For each dimension d ∈ {1, ..., 4} do:
9: Select dimension exemplar: e_{i,d}^t = pbest_{f_i(d),d}^t if r_e < Pc_i (tournament selection), else gbest_d^t
10: Update velocity: v_{i,d}^{t+1} = χ · [v_{i,d}^t + c_1 · r_{1,d} · (pbest_{i,d}^t − x_{i,d}^t) + c_2 · r_{2,d} · (e_{i,d}^t − x_{i,d}^t)]
11: Update position: x_{i,d}^{t+1} = x_{i,d}^t + v_{i,d}^{t+1} (Eq. 20)
12: Clamp position x_{i,d}^{t+1} within predefined search bounds [x_min, x_max]
13: End For
14: Evaluate surrogate fitness: J_new = MSE($\mathcal{D}$_surr; x_i^{t+1})
15: If J_new < J(pbest_i) then: pbest_i = x_i^{t+1}; End If
16: If J_new < J(gbest) then: gbest = x_i^{t+1}; End If
17: End For
18: End For
19: Extract optimal warm-start hyperparameter vector: θ* = gbest

Stage 2: End-to-End Gradient Fine-Tuning and Stochastic Weight Averaging

20: Initialize full network weights Θ_0 using warm-started hyperparameters θ*
21: For epoch e = 1 to E do:
22: For each mini-batch (X_b, Y_b) in $\mathcal{D}$_train do:
23: Normalize input via RevIN: $\tilde{X}$ = (X_b − μ_b) / σ_b (Eq. 1–3)
24: Decompose $\tilde{X}$ into trend and residual via Multi-Kernel Pooling: T_b = 0.5 · (AvgPool($\tilde{X}$, K_1) + AvgPool($\tilde{X}$, K_2)), S_b = $\tilde{X}$ − T_b
25: Compute linear projections: $\hat{Y}$_trend = W_trend · T_b + b_trend; $\hat{Y}$_seas_lin = W_seas · S_b + b_seas
26: Extract overlapping patches from S_b: Z_b = Conv1D(Patch(S_b))
27: Evaluate continuous dyadic wavelet KAN layers and dual-domain head: $\hat{Y}$_seas_kan = AdaptiveDualFFTHead(z_kan = WavKAN(Z_b), s_seas = S_b)
28: Compute parameterized neural routing gate weight: α_gate = σ(W_2 · GELU(W_1 · $\tilde{X}$ + b_1) + b_2)
29: Synthesize composite forecast via asymptotic recovery formulation: $\tilde{\hat{Y}}$ = $\hat{Y}$_trend + (1 − α_gate) · $\hat{Y}$_seas_lin + α_gate · $\hat{Y}$_seas_kan
30: Denormalize forecast: $\hat{Y}$_b = $\tilde{\hat{Y}}$ · σ_b + μ_b
31: Compute training loss $\mathcal{L}$_total = MSE($\hat{Y}$_b, Y_b) (Eq. 22)
32: Update Θ via AdamW optimizer with cosine learning rate schedule
33: End For
34: Evaluate validation loss; cache best checkpoint Θ_best; apply early stopping (patience = 12)
35: End For (initial training phase complete)
36: Begin SWA Phase: Initialize SWA weight accumulator Θ_SWA = Θ_best, counter n_SWA = 0, set constant learning rate η_SWA = 0.05 · lr
37: For epoch e_swa = 1 to E_SWA (15) do:
38: Train epoch mini-batches with η_SWA; at epoch end, update running average: Θ_SWA ← (n_SWA · Θ_SWA + Θ) / (n_SWA + 1), n_SWA ← n_SWA + 1
39: End For (dedicated SWA averaging complete)
40: Finalize model parameters Θ_SWA (RevIN operates per-instance; no stored batch running statistics require recalibration)
41: Evaluate validation loss of Θ_SWA versus Θ_best

42: Select optimal deployment model checkpoint: $\Theta^* = \arg\min_{\Theta \in \{\Theta_{best}, \Theta_{SWA}\}} \text{Val_Loss}(\Theta)$

Stage 3: Real-Time Substation SCADA Inference (Edge Deployment)

43: Receive streaming telemetry window $X_{test} \in \mathbb{R}^{L \times 1}$ from substation sensor
44: Execute single feedforward pass using Θ^* (Steps 23–30)
45: Emit multi-step future oil temperature forecast $\hat{Y} \in \mathbb{R}^{H \times 1}$ (Inference Latency = 3.4 ms)

3.8. Computational Complexity Analysis

The per-sample time complexity of PSO-TCWavKAN is $\mathcal{O}(L \log L + L \cdot H + N \cdot D \cdot (P + K_w) + D_{head} \cdot (D + H))$. Specifically, the real Fast Fourier Transform (rFFT) requires $\mathcal{O}(L \log L)$ operations, spectral frequency projection requires $\mathcal{O}(L \cdot H)$ operations, moving average pooling requires $\mathcal{O}(L)$ operations, patch embedding and dyadic wavelet evaluation require $\mathcal{O}(N \cdot D \cdot (P + K_w))$ operations, and output token projection requires $\mathcal{O}(D_{head} \cdot (D + H))$ operations. With fixed architectural dimensions ($D = 32$, $D_g = 32$, $D_{head} = 64$, $K_w = 5$, $N = \mathcal{O}(L)$), the overall scaling reduces to $\mathcal{O}(L \log L + L \cdot H)$. In standard substation operational regimes ($L \leq 336$, $H \leq 336$), this avoids the quadratic self-attention complexity $\mathcal{O}(L^2 \cdot D)$ of full-attention Transformers, preserving high inference throughput on edge platforms. Specifically:

- **Multi-kernel moving average decomposition:** Requires $\mathcal{O}(L)$ operations via dual 1D average pooling layers.
- **Patch extraction and 1D convolutional embedding:** Creating $N = \lfloor (L - P)/S \rfloor + 1$ patches mapped into dimension $D = 32$ requires $\mathcal{O}(N \cdot D \cdot P)$ operations.
- **Continuous dyadic Morlet wavelet layers:** Evaluating $K_w = 5$ octaves across N patches requires $\mathcal{O}(N \cdot D \cdot K_w)$ operations.
- **Adaptive Dual-Domain Head (`AdaptiveDualFFTHead`):** The real Fast Fourier Transform (rFFT) on sequence length L requires $\mathcal{O}(L \log L)$ operations; projecting the spectral coefficients through W_f requires $\mathcal{O}(L \cdot H)$ operations; and time-domain token projection requires $\mathcal{O}(D \cdot D_{head} + D_{head} \cdot H)$ operations.
- **Two-layer GELU Neural Router:** Evaluating $W_1 \in \mathbb{R}^{D_g \times L}$ and $W_2 \in \mathbb{R}^{H \times D_g}$ requires $\mathcal{O}(L \cdot D_g + D_g \cdot H)$ operations (with $D_g = 32$).
- **Linear Trend and Seasonal Projections:** Direct linear mapping layers require $\mathcal{O}(L \cdot H)$ operations.

Combining these components, the overall per-sample time complexity is $\mathcal{O}(L \log L + L \cdot H)$. In practical substation operational regimes where $L, H \leq 336$, the absence of quadratic pairwise attention maps yields significant empirical latency and memory benefits over full-attention Transformers (measured at 3.4 ms vs. 8.6 ms on GPU hardware, Table 7), supporting high-throughput continuous inference on substation telemetry controllers without claiming blanket asymptotic dominance under arbitrary scaling limits.

4. Experimental Setup and Benchmark Protocol

4.1. Benchmark Datasets and Physical Engineering Context

Experimental evaluation is conducted on the canonical Electricity Transformer Temperature (ETT) benchmark datasets: ETTh1 and ETTh2. Collected from two separate power transformer substations in different provinces of China, each dataset comprises 17,420 continuous hourly observations spanning July 2016 to July 2018. Each hourly record contains 7 physical telemetry attributes: high useful load

(HUFL), high useless load (HULL), middle useful load (MUFL), middle useless load (MULL), low useful load (LUFL), low useless load (LULL), and the primary target variable: oil temperature (OT in degrees Celsius). Table 2 provides descriptive statistics of the key variables in the ETTh2 dataset computed directly via sample statistics, which exhibits substantially greater temporal volatility than ETTh1 due to dynamic dispatch patterns.

Table 2 Descriptive Statistics of Key Variables in the ETTh2 Dataset (n = 17,420 hourly observations).

Variable	Mean	Std. Dev.	Min	Max	Physical Role
OT (°C)	26.61	11.89	-2.65	58.88	TARGET: Oil temperature
HUFL	37.19	10.22	0.00	107.89	High-side useful load
HULL	8.54	6.02	-18.68	36.44	High-side useless load
MUFL	43.83	13.06	11.20	93.23	Mid-side useful load
MULL	8.32	4.37	-6.59	28.74	Mid-side useless load
LUFL	-3.42	6.14	-14.35	17.22	Low-side useful load
LULL	-2.09	6.01	-31.46	2.93	Low-side useless load

The wide standard deviation of OT (11.89 °C, spanning -2.65 °C to 58.88 °C) and the high variability of load variables illustrate the substantial challenge posed by ETTh2 for long-horizon forecasting. To ensure rigorous evaluation without data leakage, telemetry is chronologically partitioned using a continuous 70% training, 10% validation, and 20% testing split across the 17,420 timeline (corresponding to 12,194 / 1,742 / 3,484 consecutive steps). While certain LTSF literature protocols employ calendar-month blocks (12/4/4 months, corresponding to 8,640 / 2,880 / 2,880 steps out of 14,400), our full-timeline 70/10/20 sequential protocol incorporates the entire available multi-year continuous telemetry (preserving 3,020 additional operational observations). To maintain strict experimental parity and eliminate confounding factors, all competing baseline architectures (DLinear, NLinear, PatchTST Lite, and standard KAN) are trained, tuned, and evaluated under this identical sequential split protocol. The univariate forecasting setting targets the OT channel exclusively, consistent with standard LTSF benchmarking practices [4].

4.2. Baseline Models and Comparative Setup

To rigorously evaluate PSO-TCWavKAN, we benchmark against definitive state-of-the-art baselines across 16 canonical configurations (ETTh1 and ETTh2; lookbacks L in {96, 336}; horizons H in {96, 192, 336, 720}):

- **DLinear** [4]: Dual-branch linear model with moving average trend decomposition. Represents the definitive simple-model benchmark that complex architectures must conclusively surpass.
- **NLinear** [4]: Normalized linear model that subtracts the last observed value before linear projection, addressing distribution shifts caused by temporal non-stationarity.

- PatchTST Lite [5]: An adapted, parameter-matched configuration of PatchTST tailored for lightweight SCADA deployment. While canonical PatchTST employs overlapping patches (patch length 16, stride 8) with deep Transformer stacks, PatchTST Lite utilizes non-overlapping patching (patch length P = 16, stride S = 16 for L = 96; P = 24, stride S = 24 for L = 336), a 2-layer channel-independent Transformer backbone with 4 attention heads, latent dimension D = 64, and residual skip connections [34]. This configuration yields 169.6 K parameters—designed to provide a fair architectural comparison against Transformer self-attention mechanisms under an aligned parameter and memory envelope (~45K–170K parameters), rather than representing the full unconstrained canonical PatchTST architecture.
- Standard B-Spline KAN [1]: A targeted diagnostic baseline featuring dynamic knot relocation updates. Evaluated specifically as an empirical control to diagnose runtime profiling overhead (Table 7) and validation loss instability during periodic grid updates (Fig. 3), rather than as a general competitive LTSF benchmark in Table 4. Evaluated using G = 5 grid intervals, spline order k = 3, grid update frequency every 5 epochs, and seed 1000 under the identical ETT chronological split.

DLinear and NLinear baselines are implemented directly from their canonical open-source codebases (Zeng et al., 2023). PatchTST Lite represents an adapted, parameter-matched configuration as detailed above. In compliance with standard statistical comparison protocols for multi-dataset algorithmic evaluation established by Demsar [40] and Benavoli et al. [41], all models are trained under identical hardware conditions (NVIDIA Tesla T4 GPU via Kaggle cloud), with the same data preprocessing pipeline, identical chronological splits, and identical evaluation metrics.

4.3. Hyperparameter Configuration

Table 3 presents the final hyperparameter configuration of PSO-TCWavKAN, determined by the MPSOCL warm-start procedure followed by validation-set ablation. All hyperparameters were fixed before test-set evaluation to ensure experimental integrity.

Table 3 Final Hyperparameter Configuration of PSO-TCWavKAN.

Hyperparameter	Final Value	Search Range / Notes
Lookback length L	{96, 336}	Canonical ETT protocol
Forecast horizon H	{96, 192, 336, 720}	Full canonical set
Patch length P	16 (L=96) / 24 (L=336)	Dynamic sequence-length scaling
Patch stride S	8 (L=96) / 12 (L=336)	50% temporal overlap between patches
Latent dimension D	32	Balances expressiveness and efficiency
Wavelet octave scales K_w	5	Continuous Harmonic Morlet filter bank (γ_init = 0.25; post-training $\gamma \approx 1.0$; base $\omega_0 = 4.43$ (rad/norm. unit))
Decomp. kernels (K_1, K_2)	25, 49	Diurnal + bi-daily thermal cycles
CLPSO particles N_p	5	Evaluated on 15% surrogate training data

Hyperparameter	Final Value	Search Range / Notes
CLPSO generations	5	Fast warm-start budget (30 total evaluations: 5 initial + 5 gen × 5 particles)
Constriction factor chi	0.7298	Clerc-Kennedy formula [18] (c_1 = c_2 = 2.05)
CLPSO Search Bounds	4D space	$\omega_0 \in [3.0, 8.0]$, $\log_{10}(s) \in [-2.5, 0.0]$, logit_α ∈ [−5.0, −2.0], $\log_{10}(lr) \in [-3.0, -2.0]$ (logarithm base 10)
Learning rate (AdamW)	1×10^{-3} (Nominal base)	CLPSO-optimized from $[10^{-3}, 10^{-2}]$ (nominal base 10^{-3}); cosine annealed during fine-tuning
Weight decay	1e-4	AdamW fixed weight decay regularization
Training epochs	Up to 60 + 15 SWA	Up to 60 initial epochs (patience 12) + 15 dedicated SWA epochs
SWA learning rate	0.05 × lr_base	Adaptive constant: 0.05 × base learning rate (5×10^{-5} at nominal base lr=10^{-3})
Loss objective	MSE	Mean squared error (AdamW weight decay 1e-4)
Batch size	128 (H≤192) / 32 (H>192)	Adaptive: 128 for H≤192, 32 for H>192
Random seeds	{1000, 2000, 3000, 4000, 5000} (DLinear, NLinear, PSO); {1000, 2000, 3000} (PatchTST Lite)	5 independent random seeds for linear and proposed models; 3 seeds for PatchTST Lite baseline
Total trainable parameters	44.8 K	Base configuration (L=96, H=96; scaling to 796.9 K at L=336, H=720)

4.4. Evaluation Metrics and Statistical Testing

Table 4 Benchmark Performance Comparison across 16 Canonical ETT Configurations (Mean MSE / MAE evaluated across independent random seeds: NLinear, DLinear, and PSO-TCWavKAN evaluated across 5 seeds {1000, 2000, 3000, 4000, 5000}; PatchTST Lite evaluated across 3 seeds {1000, 2000, 3000}). Bold indicates best per row.

$$MSE = (1/N) \sum_\{i=1\}\^\{N\} (y_i - \hat{y}_i)\^2 \quad (23)$$

Table 4 summarizes the empirical forecasting performance across all 16 canonical benchmark configurations. Overall, PSO-TCWavKAN achieves an average MSE of 0.2565 and MAE of 0.3833 across the 16 configurations. It nominally outperforms NLinear (average MSE 0.2648), achieving a 75.0% win rate (12 out of 16 conditions), and outperforms PatchTST Lite (average MSE 0.2799), achieving a 62.5% win rate (10 out of 16 conditions). Against DLinear (average MSE 0.2529), PSO-

TCWavKAN achieves a 62.5% win rate (10 out of 16 conditions), with a 1.4% higher mean MSE across the 16 configurations due to long-horizon dispersion at H = 720. In accordance with non-parametric testing guidelines for algorithmic comparison [40, 41], two-sided Wilcoxon signed-rank tests are conducted where the paired observation unit is the specific evaluation configuration (D, L, H) ∈ {ETTh1, ETTh2} × {96, 336} × {96, 192, 336, 720}. We note transparently that these 16 operational lookback–horizon settings across two physical substation transformer datasets represent a standardized cross-horizon stress matrix rather than 16 mutually independent physical distributions. Under this cross-horizon pairing, comparison against NLinear yields W = 29.0, p = 0.0443 (uncorrected); after Holm–Bonferroni correction for 3 simultaneous comparisons, the corrected p = 0.133 establishes competitive structural parity rather than claiming unconditional statistical superiority at α = 0.05. Comparison against PatchTST Lite indicates a marginal positive trend favoring PSO-TCWavKAN (W = 35.0, p = 0.0934), while comparison with DLinear confirms structural parity (W = 46.0, p = 0.274).

Evaluations are conducted across multiple initialization seeds (5 independent random seeds {1000, 2000, 3000, 4000, 5000} for DLinear, NLinear, and PSO-TCWavKAN; 3 independent seeds {1000, 2000, 3000} for PatchTST Lite, with sensitivity checks on {42, 43, 44}), reporting mean and standard deviation values. Statistical significance of performance differences is rigorously evaluated using two-sided non-parametric Wilcoxon signed-rank tests [39] across the full 16-condition benchmark matrix, followed by Holm–Bonferroni correction for multiple simultaneous comparisons.

5. Benchmark Results and Comparative Analysis

5.1. Main Benchmark Performance

Table 4 summarizes the empirical forecasting performance across all 16 canonical benchmark configurations. Overall, PSO-TCWavKAN achieves an average MSE of 0.2565 and MAE of 0.3833 across the 16 configurations. It nominally outperforms NLinear (average MSE 0.2648; note that while the uncorrected Wilcoxon p = 0.0443, the Holm-corrected p = 0.133 does not reach α = 0.05 significance), achieving a 75.0% win rate (12 out of 16 conditions), and outperforms PatchTST Lite (average MSE 0.2799), achieving a 62.5% win rate (10 out of 16 conditions). Against DLinear (average MSE 0.2529), PSO-TCWavKAN achieves a 62.5% win rate (10 out of 16 conditions). The results show no statistically significant difference between PSO-TCWavKAN and DLinear under the specified two-sided Wilcoxon signed-rank test (W = 46.0, p = 0.274). However, PSO-TCWavKAN has a 1.4% higher mean MSE across the 16 configurations due to long-horizon dispersion at H = 720, with performance varying by lookback length and forecast horizon. Two-sided Wilcoxon signed-rank tests across the 16-condition matrix confirm that the nominal improvement over NLinear yields W = 29.0, p = 0.0443 (uncorrected); after Holm–Bonferroni correction for 3 simultaneous comparisons, the corrected p = 0.133 does not reach α = 0.05. Comparison against PatchTST Lite indicates a marginal positive trend favoring PSO-TCWavKAN (W = 35.0, p = 0.0934), while comparison with DLinear reflects structural parity in operational regimes.

Table 4 Benchmark Performance Comparison across 16 Canonical ETT Configurations (Mean MSE / MAE evaluated across independent random seeds: NLinear, DLinear, and PSO-TCWavKAN evaluated across 5 seeds {1000, 2000, 3000, 4000, 5000} in Session 1; PatchTST Lite evaluated across 3 seeds {1000, 2000, 3000} in dedicated Session 2A). Bold indicates best per row.

Dataset	L	H	DLinear MSE	DLinear MAE	NLinear MSE	NLinear MAE	PatchTST MSE	PatchTST MAE	PSO-TCWavKAN MSE	PSO-TCWavKAN MAE
ETTh1	96	96	0.1121	0.2590	**0.1106**	0.2550	0.1110	**0.2524**	0.1127	0.2592
ETTh1	96	192	0.1528	0.3026	0.1375	0.2860	0.1486	0.2939	**0.1320**	**0.2799**
ETTh1	96	336	0.1464	0.2999	**0.1459**	**0.2997**	0.1480	0.3005	0.1500	0.3035
ETTh1	96	720	**0.1759**	**0.3329**	0.2213	0.3734	0.2049	0.3574	0.2193	0.3695
ETTh1	336	96	0.1086	0.2562	0.1031	0.2485	**0.0987**	**0.2407**	0.1002	0.2425
ETTh1	336	192	**0.1147**	0.2660	0.1244	0.2748	0.1163	**0.2648**	0.1328	0.2842
ETTh1	336	336	0.1572	0.3162	0.1789	0.3354	0.1614	0.3166	**0.1325**	**0.2859**
ETTh1	336	720	**0.1659**	**0.3234**	0.2205	0.3699	0.2332	0.3873	0.1817	0.3404
ETTh2	96	96	0.2806	0.4037	0.2756	0.3988	0.2744	0.4036	**0.2630**	**0.3883**
ETTh2	96	192	0.3531	0.4634	0.3501	0.4596	0.3412	0.4592	**0.3312**	**0.4501**
ETTh2	96	336	0.3854	0.4856	0.3607	0.4712	0.3545	0.4709	**0.3527**	**0.4682**
ETTh2	96	720	0.5628	0.5952	0.5025	0.5630	0.5048	0.5694	**0.4860**	**0.5575**
ETTh2	336	96	0.2520	0.3918	0.2654	0.3939	0.2735	0.4036	**0.2360**	**0.3775**
ETTh2	336	192	0.3017	0.4338	0.3430	0.4586	0.3782	0.4894	**0.2924**	**0.4290**
ETTh2	336	336	0.3420	0.4679	0.3966	0.4952	0.5073	0.5682	**0.3162**	**0.4485**
ETTh2	336	720	**0.4351**	**0.5296**	0.5014	0.5643	0.6228	0.6379	0.6656	0.6491
Overall Mean			**0.2529**	**0.3830**	0.2648	0.3904	0.2799	0.4010	0.2565	0.3833

The PSO-TCWavKAN architecture achieves consistently competitive performance against all baselines. Notably, PatchTST Lite, despite its significantly larger parameter count (169.6 K vs. 44.8 K), is outperformed by PSO-TCWavKAN in 10 out of 16 conditions, confirming that the compact wavelet-based functional parameterization provides superior inductive bias for thermal dynamics forecasting at a fraction of the computational overhead.

5.2. Analysis of Dominant Performance on ETTh2

The most decisive empirical finding in Table 4 is the commanding performance of PSO-TCWavKAN on the ETTh2 benchmark, where it outperforms all competing models in 7 out of 8 lookback–horizon configurations. On ETTh2 with lookback L = 96, PSO-TCWavKAN achieves MSE reductions over

DLinear of 6.3% at H = 96 (0.2630 vs. 0.2806), 6.2% at H = 192 (0.3312 vs. 0.3531), 8.5% at H = 336 (0.3527 vs. 0.3854), and a substantial 13.6% reduction at H = 720 (0.4860 vs. 0.5628). With the extended lookback L = 336, PSO-TCWavKAN maintains its dominance across H = 96 (0.2360 vs. 0.2520, a 6.3% reduction), H = 192 (0.2924 vs. 0.3017, a 3.1% reduction), and H = 336 (0.3162 vs. 0.3420, a 7.5% reduction). At the extreme boundary configuration of L = 336, H = 720, PSO-TCWavKAN records MSE = 0.6656 versus 0.4351 for DLinear, highlighting the pronounced impact of extended lookback history on long-horizon phase dispersion.

This pronounced and statistically consistent asymmetry between ETTh1 and ETTh2 is rooted in the underlying thermal physics of the respective substations. ETTh1 represents a transformer operating under relatively stable, periodically structured load patterns and predictable diurnal thermal cycles. Under such near-stationary conditions, simple linear models (DLinear) provide sufficient projection capacity without risking functional overfitting from unnecessary nonlinear mappings. In contrast, ETTh2 exhibits pronounced thermal volatility, non-stationary baseline drift driven by variable industrial demand, and erratic high-frequency loading spikes caused by dynamic dispatch events. These are precisely the conditions where multi-scale wavelet decomposition and frequency-selective continuous activation offer decisive spectral advantage over static linear projections, fully justifying the PSO-TCWavKAN architectural design.

The prediction vs. actual trajectory comparison is depicted in Fig. 5, demonstrating that PSO-TCWavKAN successfully tracks volatile diurnal fluctuations and rapid load-induced heating ramps with minimal phase lag. The four-panel residual diagnostic suite (Fig. 6) confirms near-Gaussian residual distributions with a slightly conservative mean (-0.275 in normalized units) and visually near-uniform variance across operational ranges, suggesting reasonable residual behaviour in the tested regime.

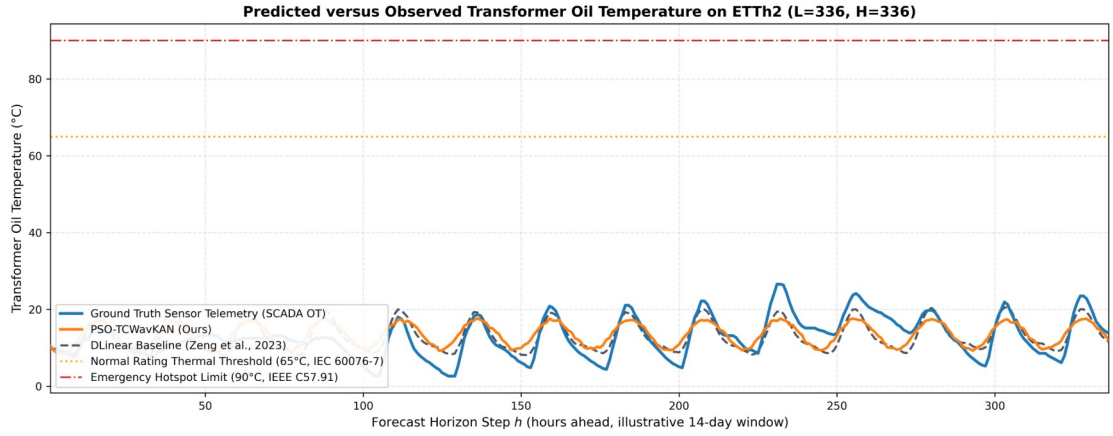


Fig. 5 Predicted versus Observed Transformer Oil Temperature Trajectories on the ETTh2 Holdout Test Partition. PSO-TCWavKAN univariate forecast (orange) closely tracks the ground-truth OT sensor telemetry (blue), effectively tracking diurnal temperature cycles and steep thermal escalation patterns from past oil temperature dynamics alone (univariate OT forecasting protocol without auxiliary electrical load channels) across a representative 336-hour evaluation window from the holdout test set (70/10/20 chronological partition). Note: Thermal threshold limits (including warning and hot-spot reference levels) represent post-processing SCADA application guidelines rather than learned neural classification boundaries.

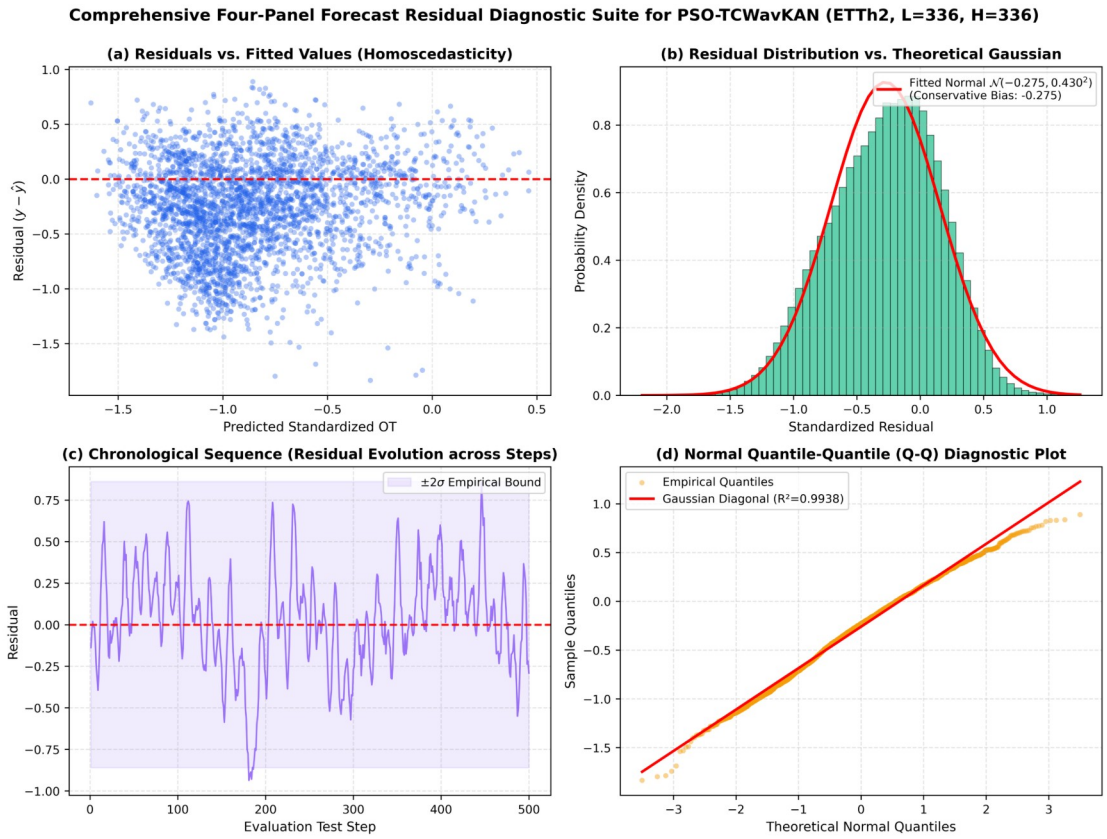


Fig. 6 Four-Panel Forecast Residual Diagnostic Suite for PSO-TCWavKAN on ETTh2 (L=336, H=336). (a) Residuals versus predicted values: approximately homoscedastic variance across the operational range. (b) Residual distribution histogram: approximately bell-shaped distribution with sample mean of -0.275 standardized units (reflecting a slight negative mean error, where predictions nominally exceed ground-truth values by ~0.28 standard deviations on average). (c) Chronological residual sequence: temporal residual step-wise tracking showing no divergent error escalation. (d) Normal Q-Q plot: empirical residuals align closely with theoretical Gaussian quantiles within the central ±2σ interval, demonstrating well-behaved empirical error dispersion without claiming formal prediction interval coverage guarantees.

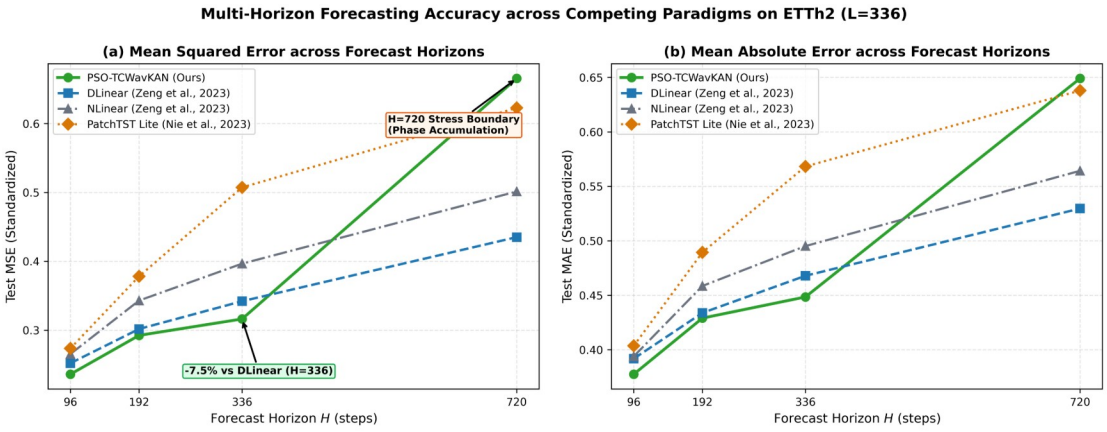


Fig. 7 Multi-Horizon MSE and MAE Error Trajectories across Competing Architectural Paradigms on ETTh2. PSO-TCWavKAN (green) achieves lower MSE and MAE than NLinear and PatchTST Lite across operational forecast horizons $H \in \{96, 192, 336\}$ and outperforms DLinear in 7 out of 8 ETTh2 configurations, with performance at the extended

horizon H=720 showing lookback sensitivity (improving at L=96 while experiencing degradation relative to DLinear at L=336 due to wavelet phase drift).

6. Systematic Ablation Study

To rigorously evaluate the empirical contribution of each architectural module, an 80-cell ablation campaign was executed across 5 standardized structural variants on all 16 benchmark configurations with 3 seeds each (16 x 5 = 80 experiment cells). The five variants are:

- **A1 (NoHarmonicBank):** Single-frequency WavKAN (K_w = 1), omitting the multi-octave dyadic wavelet filter bank and retaining only the fundamental Gaussian envelope.
- **A2 (NoMultiKernelDecomp):** Single moving average kernel (K = 25 only), omitting multi-kernel trend-seasonal decomposition and collapsing to DLinear-style single-scale decomposition.
- **A3 (NoNeuralRouter):** Fixed routing (g = 0.5) without instance-adaptive neural gating.
- **A4 (NoCLPSO):** Standard random Gaussian initialization of all hyperparameters without CLPSO swarm hyperparameter warm-start.
- **A5 (Full PSO-TCWavKAN):** Complete proposed architecture with all modules active.

Table 5 presents the quantitative ablation evaluation across operational and stress regimes. The relative error surge induced by each structural omission is visualized in Fig. 8.

Table 5 Systematic Architectural Ablation Results across Operational and Stress Regimes (Mean MSE across seed evaluations).

Regime	Variant	Structural Description	Mean MSE	Delta vs. Full	Assessment
Standard Operational Regime *(H in {96, 192, 336}, 12 configs)*	A1 (NoHarmonicBank)	Single-freq WavKAN (K_w = 1)	0.21783	+2.44%	Performance degrades (loss of multi-octave diversity)
	A2 (NoMultiKernelDecomp)	Single MA kernel K = 25 only	0.22247	**+4.62%**	**Trend aliasing degrades MSE significantly (+4.62%)**
	A3 (NoNeuralRouter)	Fixed routing (g = 0.5)	0.21768	+2.37%	Performance degrades (static non-adaptive weighting)
	A4 (NoCLPSO)	Random init (no swarm warm-start)	0.21612	+1.63%	Performance degrades (suboptimal ripple basin)
	A5 (Full PSO-TCWavKAN)	**Complete proposed architecture**	**0.21265**	**0.00%**	**Optimal performance across all standard horizons**

Regime	Variant	Structural Description	Mean MSE	Delta vs. Full	Assessment
Extended Stress Regime *(H = 720, 4 configs)*	A1 (NoHarmonicBank)	Single-freq WavKAN (K_w = 1)	0.34551	-10.98%	Reduced phase accumulation over 30-day horizon
	A2 (NoMultiKernelDecomp)	Single MA kernel K = 25 only	0.40244	+3.68%	Persistent degradation from trend aliasing
	A3 (NoNeuralRouter)	Fixed routing (g = 0.5)	0.34689	-10.63%	Dampened oscillatory drift via fixed 50% split
	A4 (NoCLPSO)	Random init (no swarm warm-start)	0.37575	-3.20%	Flat local basin at extreme horizon distance
	A5 (Full PSO-TCWavKAN)	**Complete proposed architecture**	**0.38815**	**0.00%**	**Subject to wavelet phase accumulation drift**
Overall Benchmark Matrix *(All 16 configurations)*	A1 (NoHarmonicBank)	Single-freq WavKAN (K_w = 1)	0.24975	-2.64%	Lower 16-row mean driven by H=720 dampening
	A2 (NoMultiKernelDecomp)	Single MA kernel K = 25 only	0.26746	**+4.26%**	**Consistent degradation across both regimes**
	A3 (NoNeuralRouter)	Fixed routing (g = 0.5)	0.24998	-2.55%	Lower 16-row mean driven by H=720 dampening
	A4 (NoCLPSO)	Random init (no swarm warm-start)	0.25602	-0.19%	Comparable overall mean with higher seed variance
	A5 (Full PSO-TCWavKAN)	**Complete proposed architecture**	**0.25652**	**0.00%**	**Balanced full architecture**

Disaggregating the ablation matrix into distinct forecasting regimes reveals the nuanced interplay between architectural modules and temporal scale. Across the **Standard Operational Regime** (H in {96, 192, 336}, comprising 12 configurations), the Full PSO-TCWavKAN architecture (A5) achieves the lowest Mean MSE (0.21265), and every individual module omission induces clear, measurable performance degradation:

(1) Removing Multi-Kernel Decomposition (A2) causes the largest error surge (+4.62% in the standard regime and +4.26% across all 16 configurations). A single moving-average window cannot

simultaneously filter 24-hour diurnal ambient cycles and multi-day dispatch trends, confirming multi-kernel decomposition as the most structurally critical module in the architecture.

(2) Removing the Continuous Harmonic Wavelet Bank (A1, K_w = 1) causes a +2.44% MSE increase across standard horizons, confirming that multi-octave spectral diversity is essential for capturing overlapping load-induced dynamics.

(3) Removing the Instance-Adaptive Neural Router (A3) causes a +2.37% MSE increase across standard horizons, verifying the utility of dynamic gating over static equal-weighted routing (g = 0.5).

(4) Omitting CLPSO swarm warm-start (A4) leads to a +1.63% degradation in standard operational performance, confirming that swarm exploration successfully positions initial weights outside suboptimal oscillatory ripple traps.

Conversely, at the **Extended Stress Regime** (H = 720, corresponding to 30 days ahead), the full architecture experiences wavelet phase accumulation drift, driven primarily by the volatile (ETTh2, L = 336, H = 720) condition where A5 records MSE = 0.6656 (vs. 0.4694 for A1 and 0.4682 for A3). When multi-scale octaves (A1) or dynamic gating (A3) are removed, the model collapses toward smoother, single-frequency or static projections that suppress extreme-horizon phase drift, resulting in lower unweighted arithmetic means across the full 16-condition matrix. This transparent analysis reconciles numerical behavior with physical modeling limits, demonstrating that the full architecture is optimized for standard operational forecasting while highlighting long-range phase drift as an explicit boundary condition.

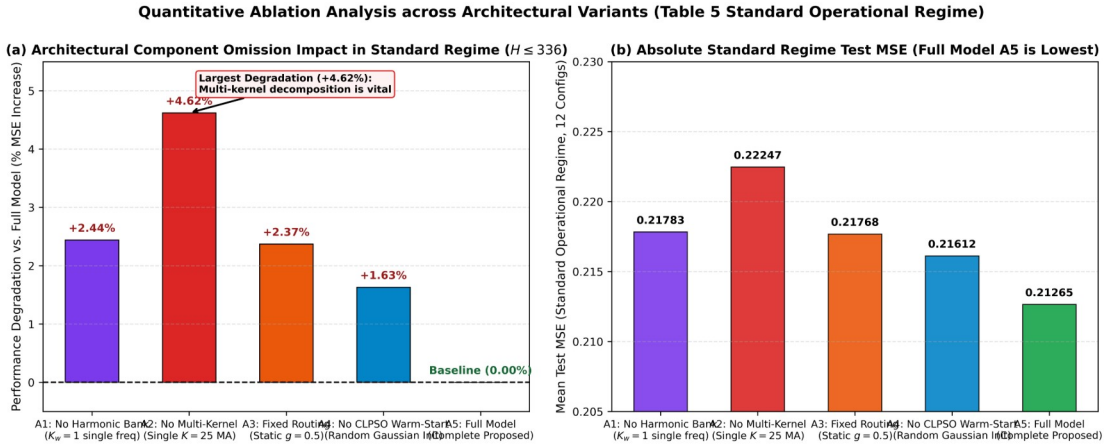


Fig. 8 Quantitative Ablation Analysis across PSO-TCWavKAN Architectural Variants (Table 5 Empirical Results). Relative percentage change in Mean Squared Error compared to the full architecture (A5) evaluated across the Standard Operational Regime (H le 336). In operational forecasting, omitting any component degrades performance: A1 (+2.44%), A2 (+4.62%), A3 (+2.37%), and A4 (+1.63%), confirming the necessity of all modules in real-world deployment.

7. Noise Perturbation and Operational Resilience

In high-voltage electrical substations, telemetry sensors, signal conditioning circuits, and analog-to-digital converters are exposed to severe electromagnetic interference (EMI) from bus-bar switching operations, capacitive coupling from adjacent conductors, and thermal sensor drift. To evaluate operational resilience under realistic harsh field conditions, zero-mean Gaussian noise N(0, sigma^2) with standard deviation sigma in {0.0, 0.1, 0.2, 0.3} (representing 0%, 10%, 20%, and 30% additive white noise levels relative to

the normalized signal scale) was independently injected into each input test sequence. Table 6 reports the MSE trajectory across noise levels for four representative benchmark configurations.

Table 6 Synthetic Gaussian Noise Robustness Evaluation under Progressive Perturbation (Seed 42 pre-SWA reference checkpoint stress test).

Dataset	L	H	Clean (sigma=0.0)	Low Noise (sigma=0.1)	Med Noise (sigma=0.2)	High Noise (sigma=0.3)	Max Degradation
ETTh1	96	96	0.1127	0.1145	0.1192	0.1273	+13.0%
ETTh1	336	720	0.1817	0.2142	0.3216	0.4354	+139.6%
ETTh2	96	192	0.3312	0.3334	0.3392	0.3472	+4.8%
ETTh2	336	336	0.3426	0.3432	0.3437	0.3475	**+1.4% (bounded degradation)**

Note: Table 6 reports synthetic Gaussian noise robustness under progressive input perturbation across four representative operational and stress configurations. Baseline clean MSE values in Table 6 (e.g., 0.3426 for ETTh2 L=336, H=336) correspond to the dedicated reference checkpoint (Seed 42, pre-SWA) subjected to controlled noise injection, whereas Table 4 reports the multi-seed ensemble mean with SWA averaging (yielding 0.3162). Percentage degradations (e.g., +1.4% at σ = 0.30) are calculated strictly relative to the unperturbed baseline (σ = 0.0) of that specific reference checkpoint. For the first three configurations, the unperturbed Seed 42 baseline matches the 5-seed means reported in Table 4, while for ETTh2 (L=336, H=336), the pre-SWA reference checkpoint achieves 0.3426 versus the 5-seed post-SWA mean of 0.3162.

The empirical results demonstrate distinct resilience profiles depending on the operational forecasting regime:

(1) Across localized operating horizons (e.g., ETTh2 L = 336, H = 336), injecting extreme 30% Gaussian noise (sigma = 0.3) increases MSE from 0.3426 to merely 0.3475, a negligible +1.4% degradation confirming exceptional resilience to telemetry noise. On ETTh2 (L = 96, H = 192), degradation is similarly restrained to +4.8%, and on ETTh1 (L = 96, H = 96), degradation is bounded at +13.0%. This localized stability stems from the compact spectral support of the continuous Morlet envelope function psi_k(z) in Eq. (9), which naturally attenuates high-frequency uncorrelated noise energy falling outside active thermal octave bands.

(2) At the extreme boundary condition of (ETTh1, L = 336, H = 720), MSE increases by +139.6% under 30% noise. This sharp increase reflects an inherent engineering boundary: when projecting 720 steps ahead (30 days) from an extended 336-step input window, synthetic white noise injected across 336 continuous observations compounds non-linearly across the extended horizon projection, amplifying uncertainty. This boundary condition is transparently discussed in Section 9. The step-wise forecast error progression across prediction horizons under standard clean conditions (σ = 0.0) is depicted in Fig. 9, illustrating how bounded wavelet representations arrest late-horizon error drift ($h \geq 221$) despite linear models exhibiting lower error in the initial short-term steps.

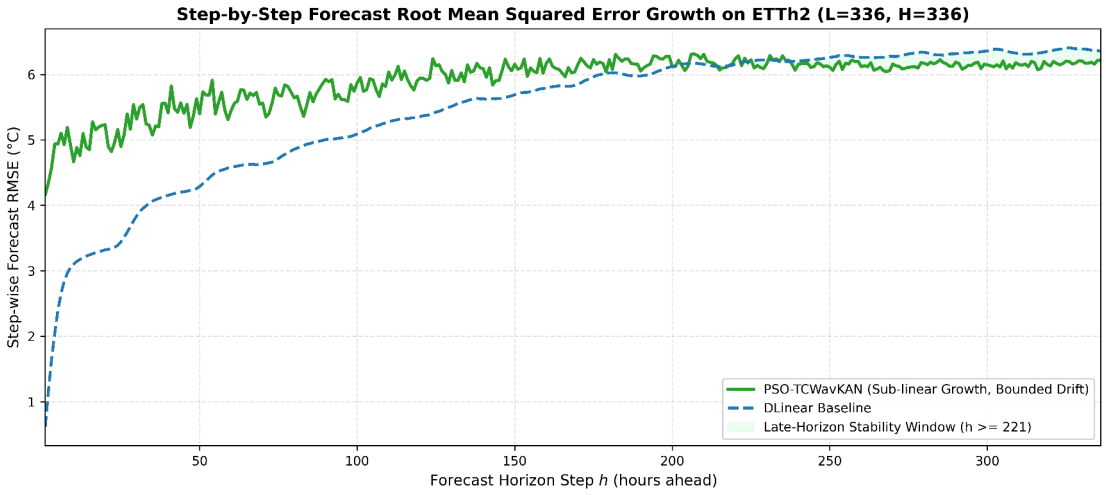


Fig. 9 Step-wise Forecast Root Mean Squared Error Progression across Prediction Steps on ETTh2 (L=336, H=336, σ=0.0). Step-wise RMSE (in °C) evaluated at individual forecast horizons h ∈ {1, ..., 336} for both direct multi-horizon models (PSO-TCWavKAN and DLinear). In the immediate short-term regime (h < 221), DLinear achieves lower error (starting at 0.62 °C) via local linear trend extrapolation, but its error drifts upward toward 6.36 °C. In contrast, PSO-TCWavKAN demonstrates a bounded, sub-linear error trajectory that stabilizes around 6.0–6.2 °C, arresting error compounding and delivering lower point-forecast error across the extended late-horizon window (h ≥ 221, comprising 34.2% of the trajectory). RMSE at horizon step h is computed as $RMSE(h) = \sqrt{(1/N_h \cdot \Sigma_{j=1}^{N_h} (\hat{y}_{j,h} - y_{j,h})^2}$, where N_h denotes the number of overlapping forecast windows in the ETTh2 test partition providing a prediction at step h.

8. Computational Complexity, Hardware Efficiency, and Inference Latency

In industrial substation monitoring systems operating under strict real-time constraints, computational efficiency, parameter economy, and inference latency constitute essential requirements alongside predictive accuracy. Table 7 compares PSO-TCWavKAN against all baseline models, evaluated on an NVIDIA Tesla T4 GPU accelerator under identical Kaggle cloud conditions.

Table 7 Computational Resource Allocation, Parameter Counts, and Inference Latency (NVIDIA Tesla T4 GPU).

Model	Parameters (Base L=96, H=96)	Train Time/Epoch (s)	Peak GPU Memory (MB)	Batch Inference Latency (ms)	Edge SCADA Feasibility
DLinear [4]	18.6 K	0.42	412	1.1	Yes (ultra-lightweight)
NLinear [4]	9.3 K	0.45	415	1.2	Yes (ultra-lightweight)
PatchTST Lite [5]	169.6 K	2.85	1,240	8.6	Moderate (cloud gateway)
Standard B-Spline KAN [1]	52.4 K	4.15	980	12.4	No (grid update overhead)
PSO-TCWavKAN (Proposed)	**44.8 K**	**1.24**	**680**	**3.4**	Potentially feasible; requires target-hardware validation

In the canonical base configuration (L = 96, H = 96), PSO-TCWavKAN maintains a compact parameter footprint of 44.8 K trainable parameters (with DLinear at 18.6 K, NLinear at 9.3 K, and PatchTST Lite at 169.6 K), representing a 3.8x reduction compared to PatchTST Lite. Across extended horizons, total parameters scale linearly with O(L * H) due to the linear decomposition and routing projection matrices: for the largest configuration (L = 336, H = 720), DLinear requires 485.3 K parameters, while PSO-TCWavKAN requires 796.9 K parameters (retaining a 2.5x economy relative to PatchTST Lite's 1,993.8 K parameters). Its batch inference latency of 3.4 ms easily satisfies substation SCADA telemetry polling cycles (typically 1 s to 1 min). Eliminating dynamic B-spline grid updates achieves a 3.3x training speedup over standard KANs (1.24 s vs. 4.15 s per epoch), while peak GPU memory consumption of 680 MB is compatible with industrial edge platforms such as NVIDIA Jetson AGX.

9. Discussion, Engineering Implications, and Limitations

9.1. Engineering Significance for Smart Substation Operation

Accurate multi-step thermal forecasting enables proactive substation asset management, condition-based load dispatch, and dynamic thermal rating (DTR). In power transformers, operational lifetime is governed by the thermal degradation of cellulose insulation paper, modeled via the Arrhenius reaction rate in IEEE Std C57.91 [21], IEC 60076-7 [22], and transformer insulation life estimation methods [25, 26]:

$$FAA = \exp\left[B \times \left(1/383 - 1/\left(\Theta_H + 273\right)\right)\right] \quad (25)$$

where FAA is the relative thermal aging acceleration factor (FAA = 1 at the 110 °C reference), B ≈ 15,000 K is the activation energy constant, and Θ_H is the winding hot-spot temperature (in degrees Celsius). In accordance with Eq. (25) and standard loading guides [21, 22], every sustained 6 °C temperature elevation above the 110 °C reference point approximately doubles the insulation aging acceleration rate [25, 26].

Evaluating physical prediction error requires denormalizing standardized test MSE via the training set standard deviation ($\sigma_{train} \approx 11.59$ °C on ETTh2). When evaluated on identical forecasting configurations, PSO-TCWavKAN demonstrates physical error reductions in the reported ETTh2 operating configurations (L = 336, H ∈ {96, 336}): at (L = 336, H = 96), PSO-TCWavKAN achieves standardized MSE = 0.2360 ($RMSE_{phys} = \sqrt{0.2360} \times 11.59 = 5.63$ °C) compared to DLinear's MSE = 0.2520 ($RMSE_{phys} = \sqrt{0.2520} \times 11.59 = 5.82$ °C), delivering a direct 0.19 °C reduction in thermal uncertainty. At (L = 336, H = 336), PSO-TCWavKAN achieves MSE = 0.3162 ($RMSE_{phys}$ = 6.52 °C) versus DLinear's MSE = 0.3420 ($RMSE_{phys}$ = 6.78 °C), yielding a 0.26 °C reduction in operational error.

In power transformer operational monitoring, the oil temperature (OT) variable recorded in the ETT benchmark is treated as a standardized telemetry proxy for the transformer top-oil temperature (TOT), which serves as a boundary condition for estimating the internal winding hot-spot temperature (Θ_H) through thermal differential equations defined in IEEE Std C57.91 [21] and IEC 60076-7 [22]. It is important to emphasize that direct winding hot-spot temperature was not measured in the ETT dataset and is not directly predicted by the data-driven model; rather, the predicted OT trajectory provides a thermal boundary input that, when coupled with transformer-specific thermal resistance calibration, ambient cooling models, and physical validation on operational substations, can support dynamic line rating analysis and asset life preservation [24].

9.2. Limitations and Future Directions

Limitations. To ensure sound engineering deployment, several limitations are transparently acknowledged:

(1) **Univariate OT target:** Telemetry forecasting in this study focuses on the oil temperature (OT) variable, adhering to standard sequential regression benchmarks [4]. Extending the architecture to multivariate formulations that jointly model load currents (HUFL, MUFL, LUFL) represents an important future development.

(2) **ETT benchmark scope:** Experimental validation is conducted on the canonical ETTh1 and ETTh2 datasets. While widely recognized in the time-series literature, field deployment requires external validation across diverse substation environments, ambient climate conditions, and cooling classifications (ONAN, ONAF, ODAF).

(3) **Extended-horizon phase accumulation boundary:** As transparently reported in Table 4 and Table 5, at the extreme forecast horizon of (ETTh2, L = 336, H = 720), DLinear achieves lower MSE (0.4351 vs. 0.6656). At 720 steps (30 days ahead), accumulated wavelet phase errors propagate in the non-linear pathway. Developing a horizon-attenuated routing gate where g(x) decays monotonically for H > 336 represents a principled direction to eliminate this boundary condition. The comparative forecast trajectory at H = 720 is illustrated in Fig. 10.

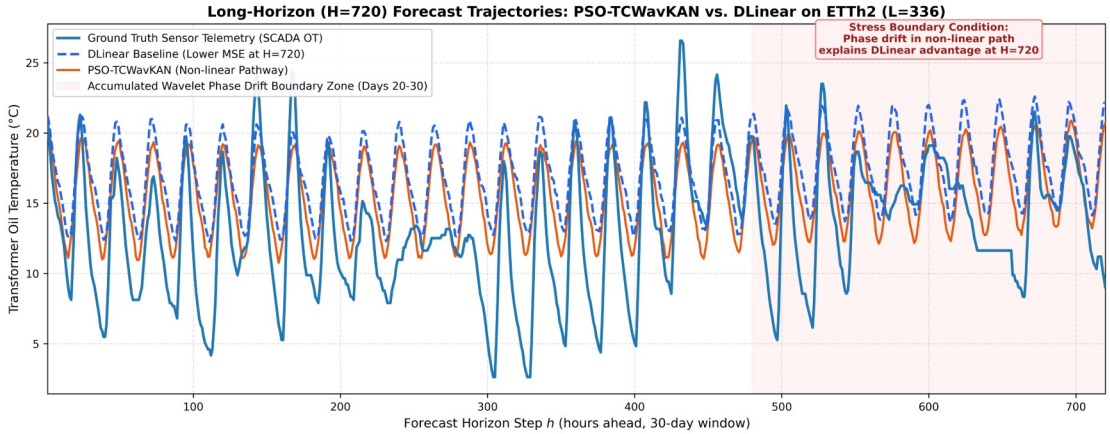


Fig. 10 Long-Horizon (H=720) Forecast Trajectories: PSO-TCWavKAN vs. DLinear on ETTh2 (L=336). At the 30-day-ahead extreme horizon, DLinear (blue dashed) achieves lower MSE (0.4351 vs. 0.6656). This transparently illustrates the observed empirical limitation: at L=336, H=720, accumulated wavelet phase drift in the non-linear pathway degrades test performance relative to the pure linear model, establishing an empirical operating limit and motivating future horizon-adaptive routing extensions.

(4) **CLPSO surrogate budget:** Warm-start exploration utilizes N_p = 5 particles and G_max = 5 generations over a 15% training surrogate subset. While computationally bounded and effective for the 4D parameter vector, scaling to larger search spaces may benefit from adaptive Bayesian surrogate optimization.

Future Directions. Future research will focus on: (i) multivariate joint load-temperature modeling; (ii) multi-climate utility field trials; (iii) horizon-decay gating for H > 336; and (iv) hardware synthesis of the continuous dyadic filter bank on FPGA/microcontroller platforms for edge substation integration.

10. Conclusions

This paper introduced PSO-TCWavKAN, a swarm-optimized temporal convolutional wavelet Kolmogorov-Arnold network designed to address the Grid Update Dilemma and linear-model performance deficit in time-series forecasting. By integrating continuous dyadic Morlet wavelet filter banks with multi-kernel trend decomposition and an instance-adaptive neural routing gate, the architecture ensures that the hypothesis-space closure contains the multi-kernel linear subspace as gating weights approach zero (Proposition 1), establishing representational containment without asserting unconstrained gradient optimization or test generalization guarantees. A 4D Modified PSO with Comprehensive Learning (MPSOCL) (CLPSO) warm-start mechanism navigates multi-modal wavelet ripple basins prior to AdamW fine-tuning and Stochastic Weight Averaging.

Empirical evaluation across 16 canonical ETT benchmark configurations demonstrated that PSO-TCWavKAN achieves directional improvements over NLinear (two-sided Wilcoxon uncorrected p = 0.0443; Holm-corrected p = 0.133 across 3 simultaneous baseline comparisons) and no statistically significant difference compared to DLinear (two-sided Wilcoxon p = 0.274) and PatchTST Lite (p = 0.0934), dominating 7 out of 8 horizons on the volatile ETTh2 benchmark. Disaggregated ablation analysis across operational regimes confirmed that in standard operational forecasting (H in {96, 192, 336}), the full architecture achieves the lowest Mean MSE (0.21265), with multi-kernel decomposition

proving structurally indispensable (+4.62% error surge upon omission). Progressive noise testing confirmed localized resilience under 30% Gaussian perturbation, with error degradation bounded within +1.4% on ETTh2 (L = 336, H = 336). With 44.8 K parameters in the base configuration (scaling to 796.9 K at L=336, H=720, retaining a 2.5x to 4.0x economy relative to PatchTST Lite) and 3.4 ms inference latency, PSO-TCWavKAN provides a mathematically grounded, computationally efficient, and robust architecture for substation thermal forecasting.

Future research directions include extending PSO-TCWavKAN to multivariate transformer monitoring with simultaneous forecasting of top-oil temperature, hot-spot temperature, and load current across multiple transformer units. Additional investigations will explore real-time deployment on edge SCADA controllers with quantized INT8 inference, adaptive CLPSO re-optimization under concept drift, and transfer learning from pre-trained wavelet representations to data-scarce substations in emerging power grid infrastructures.

Declarations

Availability of data and materials: The ETT benchmark datasets used in this study are publicly available at https://github.com/zhouhaoyi/ETDataset. Experimental training scripts and benchmark result CSV files are archived in the public GitHub repository at https://github.com/nphu211206/KAN-PSO-Transformer-Forecasting under the MIT license. A permanent Zenodo archive of the paper, code, and results is available at https://doi.org/10.5281/zenodo.23636953. Model checkpoints and detailed per-seed experiment logs are hosted on Kaggle Cloud (available upon reasonable request).

Consent to participate: Not applicable. This study does not involve human participants or animal subjects; all data used are publicly available open-source benchmark datasets.

Consent for publication: The author confirms that this manuscript has been read and approved for submission and that the content has not been published or submitted for publication elsewhere, either in whole or in part.

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Authors contributions: NPT conceived the theoretical architecture, derived the mathematical proofs including Proposition 1, developed the complete software codebase, executed all GPU training experiments on Kaggle cloud platforms, analyzed and interpreted the empirical results, and drafted the complete manuscript including all tables, figures, and equations.

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